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(Last Updated October 28, 2026)

Lecture 1

Random Variables

1.1 Bogleheads Own the Entire Market

- John Bogle popularized the index fund and a simple investment philosophy: "Own the entire market"
- Followers of this philosophy affectionately call themselves "Bogleheads."
- The true believers lifestyle brand is "VT and Chill"
- VT is the Vanguard Total World Stock ETF:
  - A single fund designed to approximate the global investable stock market
  - Covers 10,000+ US and International stocks
  - The basic idea: stop trying to pick winners; buy the haystack

1.2 VT Market Returns

- VT returns come from the stocks it holds:
  - Price changes, weighted by holdings
  - Dividends, distributed quarterly
- Total return combines price growth and dividends.
- Raw prices on the exchange (open, close, high, low)
- Adjusted prices account for dividends.
  - Last year: \$75 price, \$2 dividend
  - Today: \$80 price
  - The dividend makes it as if you bought at \$73
  - Total Return: $100 \times (80 - 73)/73 \approx 9.6\%$

1.3 Quantmod Yahoo Data

```
suppressMessages(library(dplyr))
suppressMessages(library(quantmod))
suppressMessages(library(lubridate))

yahoo_VT = getSymbols('VT', auto.assign = FALSE)
nrow(yahoo_VT)
```

```
## [1] 4552
```

```
tail(yahoo_VT,5)
```

```
##            VT.Open VT.High VT.Low VT.Close VT.Volume
## 2026-07-27  155.48  155.74 153.68   154.45   4989400
## 2026-07-28  153.88  154.82 153.24   154.35   2579800
## 2026-07-29  153.98  154.98 152.13   152.22   4036100
## 2026-07-30  153.98  155.57 153.77   155.45   2716700
## 2026-07-31  155.98  156.25 154.25   155.86   3216200
##            VT.Adjusted
## 2026-07-27      154.45
## 2026-07-28      154.35
## 2026-07-29      152.22
## 2026-07-30      155.45
## 2026-07-31      155.86
```

1.4 VT Price Data

```
class(yahoo_VT)
```

```
## [1] "xts" "zoo"
```

```
VT = yahoo_VT %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date)) %>%
  rename(price = 'VT.Adjusted') %>%
  select(date, price) %>%
  arrange(date) %>%
  mutate(lag_price = lag(price),
         ror_d = (price/lag_price) - 1)

class(VT)
```

```
## [1] "data.frame"
```

1.5 Worst Daily Return

```
VT %>%
  arrange(ror_d) %>%
  slice_head(n = 15)
```

```
##          date     price lag_price       ror_d
## 1  2008-10-15  22.79921  25.94887 -0.12137967
## 2  2020-03-16  50.69162  57.37014 -0.11641109
## 3  2020-03-12  53.13776  59.19154 -0.10227435
## 4  2008-12-01  19.74520  21.51475 -0.08224834
## 5  2020-03-09  59.56990  64.67337 -0.07891146
## 6  2011-08-08  29.70956  32.07265 -0.07367942
## 7  2009-01-20  19.57765  21.13176 -0.07354386
## 8  2008-10-22  21.50108  23.20231 -0.07332163
## 9  2008-11-20  18.18744  19.47874 -0.06629278
## 10 2008-11-19  19.47874  20.85884 -0.06616407
## 11 2020-03-18  50.19006  53.60412 -0.06369016
## 12 2008-09-29  28.96873  30.92957 -0.06339707
## 13 2008-10-21  23.20231  24.73273 -0.06187835
## 14 2008-11-14  20.88618  22.21163 -0.05967374
## 15 2025-04-04 103.38191 109.93499 -0.05960866
```

1.6 Random Variables

- View the daily return of VT as a random variable, x.
- A random variable is characterized by its cumulative distribution function (CDF):

$$F(a) = \Pr(x \leq a).$$

- The CDF answers: what fraction of the distribution lies below a?
- Example:

$$F(0) = \Pr(x \leq 0)$$

  is the probability of a non-positive return.
- In the data, estimate this probability with

`mean(VT$ror_d <= 0)`.

1.7 Empirical CDF

```r
mean(VT$ror_d<=0, na.rm = TRUE)
```

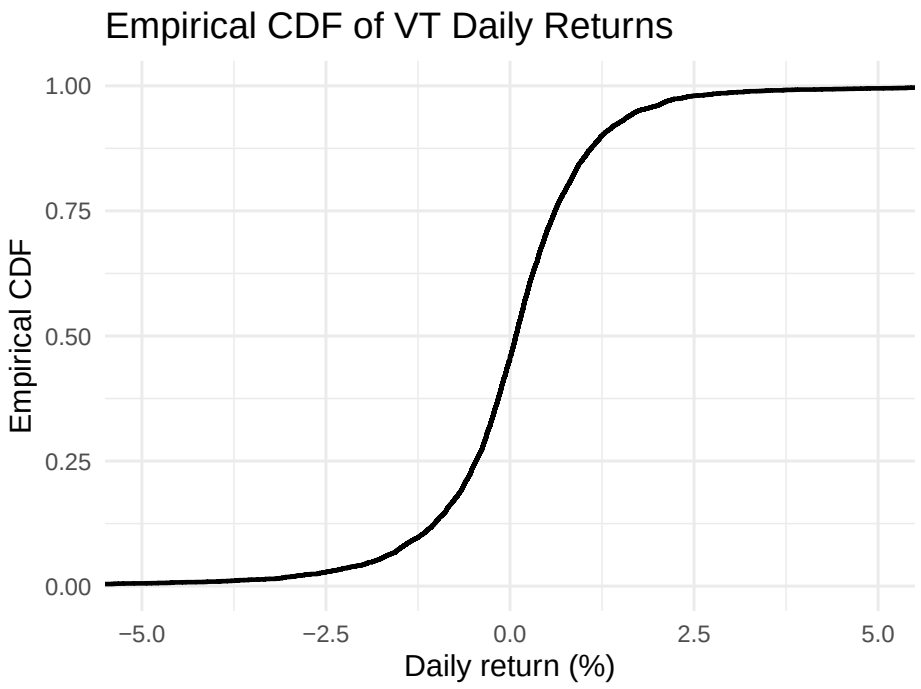
```
## [1] 0.4594595
```

```r
ecdf_df = data.frame(x = VT$ror_d) %>%
  filter(!is.na(x)) %>%
  arrange(x) %>%
  mutate(p = row_number()/n())
head(ecdf_df, 5)
```

```
##             x           p
## 1 -0.12137967 0.0002197319
## 2 -0.11641109 0.0004394639
## 3 -0.10227435 0.0006591958
## 4 -0.08224834 0.0008789277
## 5 -0.07891146 0.0010986596
```

```r
library(ggplot2)
ecdf_plot = ggplot(ecdf_df, aes(x = 100*x, y = p)) +
  geom_step(linewidth = 1) +
  coord_cartesian(xlim = c(-5, 5)) +
  labs(x = "Daily return (%)",
       y = "Empirical CDF",
       title = "Empirical CDF of VT Daily Returns") +
  theme_minimal(base_size = 14)
```

1.8



1.9 Quantile Function

- Reversing the CDF gives the quantile function.
- The 0.1-quantile, $q_{0.1}$, satisfies
$$\Pr(x \leq q_{0.1}) = 0.1.$$
That is, 10% of the distribution lies at or below $q_{0.1}$.
  - In R:
$$\texttt{quantile(VT\$ror_d, probs = 0.1)}.$$

1.10 Summarizing a Distribution with Quantiles

- Using quantiles to summarize location, spread, tails
  - In data, use sample quantiles.
- Median, $q_{0.5}$: half the values are below it.

```
quantile(VT$ror_d, probs = 0.5)
median(VT$ror_d)
```

- Minimum and maximum summarize the endpoints.

```
min(VT$ror_d),  max(VT$ror_d)
range(VT$ror_d)
```

- The interquartile range summarizes spread:
$$\text{IQR} = q_{0.75} - q_{0.25}.$$
  - It measures the width of the middle 50%.

1.11 Quantiles

```
quantile(VT$ror_d, probs = c(0.1, 0.5, 0.9), na.rm = TRUE)
```

```
##           10%           50%           90%
## -0.0122397543  0.0007971661  0.0124800276
```

```
median(VT$ror_d, na.rm = TRUE)
```

```
## [1] 0.0007971661
```

```
c(min(VT$ror_d, na.rm = TRUE), max(VT$ror_d, na.rm = TRUE))
```

```
## [1] -0.1213797  0.1345332
```

```
range(VT$ror_d, na.rm = TRUE)
```

```
## [1] -0.1213797  0.1345332
```

```
quantile(VT$ror_d,
         probs = c(0.25, 0.75),
         na.rm = TRUE,
         names = FALSE) %>%
  diff()
```

```
## [1] 0.01074818
```

```
IQR(VT$ror_d, na.rm = TRUE)
```

```
## [1] 0.01074818
```

1.12 Parametric Distributions

- So far, we have described returns empirically:
  - empirical CDF, quantiles, median, range, IQR
- Another approach is to assume x follows a parametric distribution:

$$x \sim F_\theta.$$

- The parameter vector θ controls the shape of the distribution.
- Example: normal distribution, $\theta = \{\mu, \sigma^2\}$
  - Defined by mean μ and variance σ^2
  - μ controls location; σ controls scale

1.13 Assuming VT is Normally Distributed

- Assume VT daily returns are normally distributed:

$$x \sim N(\mu, \sigma^2), \qquad \mu = 0.00042, \quad \sigma = 0.012.$$

- The normal CDF is

$$F(a) = \Pr(x \leq a \mid \mu = 0.00042, \sigma = 0.012).$$

```
pnorm(a, mean = 0.00042, sd = 0.012)
```

- The quantile function satisfies

$$\Pr(x \leq q_\alpha \mid \mu = 0.00042, \sigma = 0.012) = \alpha.$$

`qnorm(`α`, mean = 0.00042, sd = 0.012)`

1.14 Comparing Returns to a Normal Distribution

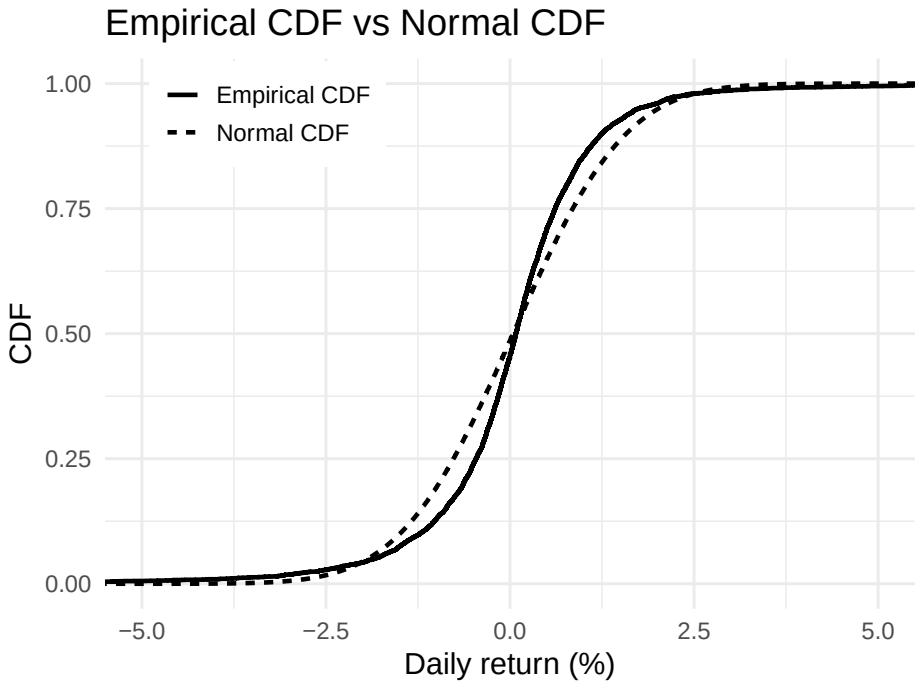
```
ecdf_df2 = ecdf_df %>%
  mutate(normal_cdf = pnorm(x, mean = 0.00042, sd = 0.012))

head(ecdf_df2, 5)
```

```
##             x            p   normal_cdf
## 1 -0.12137967 0.0002197319 1.657277e-24
## 2 -0.11641109 0.0004394639 1.059413e-22
## 3 -0.10227435 0.0006591958 5.748974e-18
## 4 -0.08224834 0.0008789277 2.808741e-12
## 5 -0.07891146 0.0010986596 1.909247e-11
```

```
ecdf_plot2 = ggplot(ecdf_df2, aes(x = 100*x)) +
  geom_step(aes(y = p, linetype = "Empirical CDF"), linewidth = 1) +
  geom_line(aes(y = normal_cdf, linetype = "Normal CDF"), linewidth = 1
  coord_cartesian(xlim = c(-5, 5)) +
  labs(x = "Daily return (%)",
       y = "CDF",
       title = "Empirical CDF vs Normal CDF",
       linetype = NULL) +
  theme_minimal(base_size = 14) +
  theme(legend.position = c(0.2, 0.9),
        legend.background = element_rect(fill = "white", color = NA))
```

1.15



1.16 Common Examples of Parametric Distributions

- Normal: useful for modeling many variables and sampling distributions

  `pnorm`, `qnorm`

- Student-t: useful for small-sample hypothesis tests and fat tails

  `pt`, `qt`

- χ^2: useful for sampling distributions and hypothesis tests

  `pchisq`, `qchisq`

- F: useful for small-sample tests of multiple restrictions

  `pf`, `qf`

Lecture 2

Expected Value

2.1 Review

- We looked at daily return of VT as a random variable, x.
- Introduced a random variables cumulative distribution function (CDF):

$$F(a) = \Pr(x \leq a).$$

  The CDF answers: what fraction of the distribution lies below a?
- Reversing the CDF gives the quantile function.
- The 0.1-quantile, $q_{0.1}$, satisfies

$$\Pr(x \leq q_{0.1}) = 0.1.$$

  That is, 10% of the distribution lies at or below $q_{0.1}$.

2.2 VT Return Data

```
VT %>%
  slice_tail(n = 15)
```

```
##          date  price lag_price         ror_d
## 1  2026-08-10 160.97    161.30 -0.0020458886
## 2  2026-08-11 160.82    160.97 -0.0009318127
## 3  2026-08-12 161.53    160.82  0.0044148204
## 4  2026-08-13 162.39    161.53  0.0053240922
## 5  2026-08-14 162.25    162.39 -0.0008621183
## 6  2026-08-17 161.83    162.25 -0.0025885866
## 7  2026-08-18 160.06    161.83 -0.0109374297
## 8  2026-08-19 160.70    160.06  0.0039984968
## 9  2026-08-20 159.93    160.70 -0.0047915637
## 10 2026-08-21 160.77    159.93  0.0052523706
## 11 2026-08-24 160.10    160.77 -0.0041674327
## 12 2026-08-25 160.99    160.10  0.0055590216
## 13 2026-08-26 160.87    160.99 -0.0007454523
## 14 2026-08-27 161.57    160.87  0.0043514156
## 15 2026-08-28 161.01    161.57 -0.0034660691
```

2.3 Probability Density Function

- The CDF gives cumulative probability:

$$F(a) = \Pr(x \leq a).$$

- The PDF is the slope of the CDF:

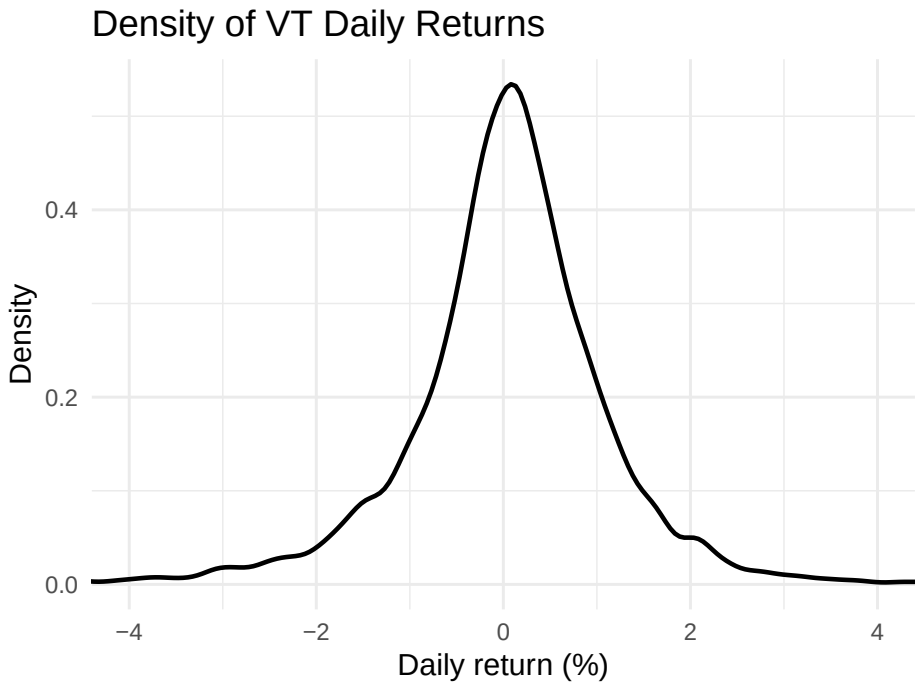
$$f(a) = \frac{dF(a)}{da}.$$

  – High density means probability is concentrated close to a.

- Probabilities are areas under the PDF:

$$\Pr(a \leq x \leq b) = \int_a^b f(t)\,dt = F(b) - F(a).$$

2.4 Density Plot



2.5 Expected Value

- The expected value is the probability-weighted average of a random variable:

$$\mathrm{E}(x) = \int x f(x)\,dx.$$

- It is sometimes called the "typical value," but this can be misleading.
- With n observations, we estimate it using the sample mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

- In R: `mean(VT$ror_d, na.rm = TRUE)`.

2.6 Properties of Expected Value

- Expected value is linear.

- Adding a constant shifts the expected value:

$$\mathrm{E}(x + a) = \mathrm{E}(x) + a.$$

- Multiplying by a constant scales the expected value:

$$\mathrm{E}(ax) = a\,\mathrm{E}(x).$$

- For two random variables:

$$\mathrm{E}(x + y) = \mathrm{E}(x) + \mathrm{E}(y).$$

- More generally:

$$\mathrm{E}(a + bx + cy) = a + b\,\mathrm{E}(x) + c\,\mathrm{E}(y).$$

2.7 Example: Expected Two-Day Return

- Let x_t be today's VT return and x_{t+1} be tomorrow's return.
- The two-day return is

$$R_{2\mathrm{d}} = (1 + x_t)(1 + x_{t+1}) - 1.$$

- Expanding,

$$R_{2\mathrm{d}} = x_t + x_{t+1} + x_t x_{t+1}.$$

- Therefore, the expected two-day return is

$$\mathrm{E}(R_{2\mathrm{d}}) = \mathrm{E}(x_t) + \mathrm{E}(x_{t+1}) + \mathrm{E}(x_t x_{t+1}).$$

- The term $\mathrm{E}(x_t x_{t+1})$ captures the *dependence* of returns across days.

2.8 Linear Approximation to Two-Day Return

- The expected two-day return is

$$\mathrm{E}(R_{2\mathrm{d}}) = \mathrm{E}(x_t) + \mathrm{E}(x_{t+1}) + \mathrm{E}(x_t x_{t+1}).$$

- One approach to approximate this expected value is to drop the product term $\mathrm{E}(x_t x_{t+1})$
- A linear approximation would be

$$\mathrm{E}(R_{2\mathrm{d}}) \approx \mathrm{E}(x_t) + \mathrm{E}(x_{t+1})$$

- If x_t and x_{t+1} come from the same distribution then

$$\mathrm{E}(x_t) = \mathrm{E}(x_{t+1}) = \mathrm{E}(x).$$

  and the expected two-day return simplifies to

$$\mathrm{E}(R_{2\mathrm{d}}) \approx 2\,\mathrm{E}(x)$$

2.9 Modeling Dependence

- There are many ways to describe how one random variable relates to another.
  - These are different notions of *dependence*.
- There are also different ways to describe how one random variable does *not* relate to another.
  - These are different notions of *independence*.
- There are weak forms of independence and stronger forms.
- These distinctions matter because they shape how we analyze data and interpret relationships between variables.

2.10 Covariance

- Covariance measures linear co-movement between two random variables.
- Definition:
$$\mathrm{Cov}(x, y) = \mathrm{E}\left[(x - \mathrm{E}(x))(y - \mathrm{E}(y))\right].$$
  - Positive: deviations tend to have the same sign.
  - Negative: deviations tend to have opposite signs.
  - Zero: no systematic linear co-movement.
- Equivalent formula:
$$\mathrm{Cov}(x, y) = \mathrm{E}(xy) - \mathrm{E}(x)\,\mathrm{E}(y).$$

2.11 A Weak Form of No Dependence

- A weak form of no dependence is zero covariance:
$$\mathrm{Cov}(x, y) = 0.$$
- Since
$$\mathrm{Cov}(x, y) = \mathrm{E}(xy) - \mathrm{E}(x)\,\mathrm{E}(y),$$
zero covariance implies
$$\mathrm{E}(xy) = \mathrm{E}(x)\,\mathrm{E}(y).$$
- So
$$\mathrm{E}(xy) = \mathrm{E}(x)\,\mathrm{E}(y) \quad \Leftrightarrow \quad \mathrm{Cov}(x, y) = 0.$$
- Zero covariance rules out linear co-movement, but it does not rule out all forms of dependence.

2.12 Expected Two-Day Return

- The expected two-day return is

$$\mathrm{E}(R_{2\mathrm{d}}) = \mathrm{E}(x_t) + \mathrm{E}(x_{t+1}) + \mathrm{E}(x_t x_{t+1}).$$

- Under zero covariance,

$$\mathrm{Cov}(x_t, x_{t+1}) = 0 \quad \Rightarrow \quad \mathrm{E}(x_t x_{t+1}) = \mathrm{E}(x_t)\,\mathrm{E}(x_{t+1}).$$

- Then the expected two-day return simplifies to

$$\mathrm{E}(R_{2\mathrm{d}}) = \mathrm{E}(x_t) + \mathrm{E}(x_{t+1}) + \mathrm{E}(x_t)\,\mathrm{E}(x_{t+1}).$$

  - It only depends on the expected daily returns

2.13 Constructing $\mathrm{E}(R_{2\mathrm{d}})$ from One-Day Returns

- If both daily returns have the same distribution,

$$x_t \sim F, \qquad x_{t+1} \sim F,$$

  then

$$\mathrm{E}(x_t) = \mathrm{E}(x_{t+1}) = \mathrm{E}(x).$$

- Under the assumption $\mathrm{Cov}(x_t, x_{t+1}) = 0$:

$$\begin{aligned}\mathrm{E}(R_{2\mathrm{d}}) &= \mathrm{E}(x_t) + \mathrm{E}(x_{t+1}) + \mathrm{E}(x_t)\,\mathrm{E}(x_{t+1}) \\ &= \mathrm{E}(x) + \mathrm{E}(x) + \mathrm{E}(x)\,\mathrm{E}(x) \\ &= 2\,\mathrm{E}(x) + \mathrm{E}(x)^2.\end{aligned}$$

- Therefore, we can construct the expected two-day return from the distribution of one-day returns:

$$\mathrm{E}(R_{2\mathrm{d}}) = (1 + \mathrm{E}(x))^2 - 1.$$

2.14 Expected Two-day Return

```
VT = VT %>%
  arrange(date) %>%
  mutate(return_one_day_ago = lag(ror_d),
         price_two_days_ago = lag(price, n = 2),
         return_2d = price / price_two_days_ago - 1)

cov(VT$ror_d, VT$return_one_day_ago, use = "complete.obs")
```

```
## [1] -1.779857e-05
```

```
mean(VT$return_2d, na.rm = TRUE)
```

```
## [1] 0.0008328586
```

```
ave_return = mean(VT$ror_d, na.rm = TRUE)
ave_return
```

```
## [1] 0.0004245876
```

```
(1 + ave_return)^2 - 1
```

```
## [1] 0.0008493554
```

2.15 Annual Returns from One-Day Returns

- There are 252 trading days in a year.
- An annual return constructed from one-day returns is

$$R_{252\text{d}} = \prod_{s=0}^{251}(1 + x_{t+s}) - 1.$$

- Therefore, the expected annual return is

$$\text{E}(R_{252\text{d}}) = \text{E}\left[\prod_{s=0}^{251}(1 + x_{t+s})\right] - 1.$$

- Expanding the product creates many expected product terms:

$$\text{E}(x_t x_{t+1}), \quad \text{E}(x_t x_{t+1} x_{t+2}), \quad \text{E}(x_t x_{t+1} x_{t+2} x_{t+3}), \quad \ldots$$

2.16 Annual Returns from One-Day Returns

- The expected annual return is

$$\text{E}(R_{252\text{d}}) = \text{E}\left[\prod_{s=0}^{251}(1 + x_{t+s})\right] - 1.$$

- The many expected product terms in this expression capture dependence across multiple days.

- Suppose all daily returns come from the same distribution:

$$x_{t+s} \sim F, \qquad s = 0, 1, 2, \ldots, 251.$$

- What assumption about no dependence would simplify the expected annual return to

$$\mathrm{E}(R_{252\mathrm{d}}) = (1 + \mathrm{E}(x))^{252} - 1.$$

2.17 Conditional Distributions

- Assume x_t and x_{t+1} have the same density function:

$$x_t \sim f, \qquad x_{t+1} \sim f.$$

- Question: what is the distribution of tomorrow's return after a bad day?

$$x_t < -0.01.$$

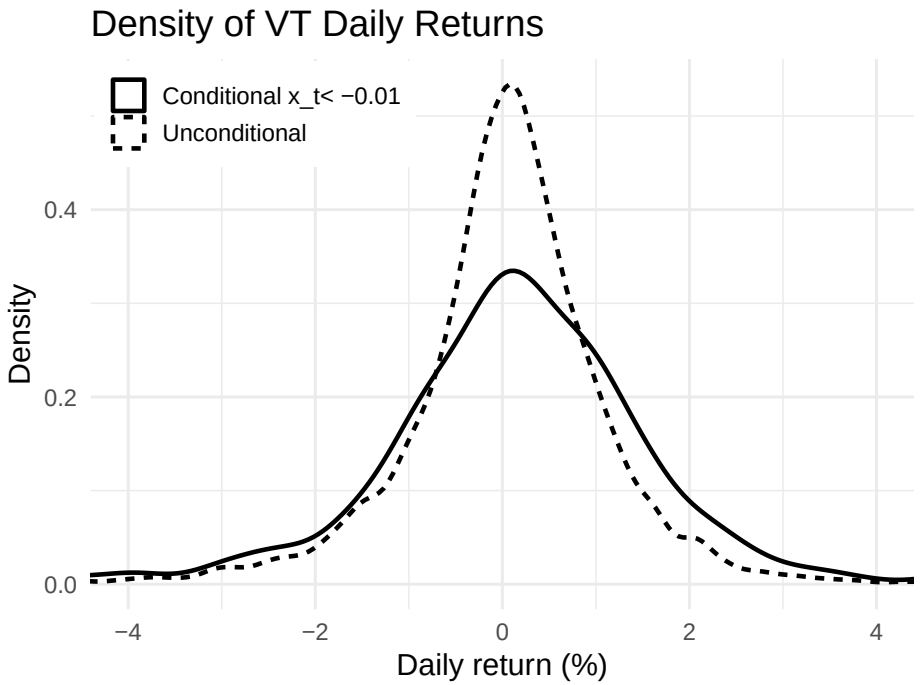
- We represent this with the conditional density:

$$f(x_{t+1} \mid x_t < -0.01).$$

- Compare the unconditional and conditional densities:

$$f(x) \qquad \text{vs.} \qquad f(x_{t+1} \mid x_t < -0.01).$$

2.18 Comparing the Distributions



2.19 Types of Dependence

- For random variables x and y, if for all values of x,
$$f(y \mid x) = f(y),$$
then y is *independent* of x.
  - Conditioning on x does not change the distribution of y.
- If, for all values of x,
$$\mathrm{E}(y \mid x) = \mathrm{E}(y),$$
then y is *mean independent* of x.
  - Conditioning on x does not change the mean of y.
- Independence implies mean independence, but mean independence does not imply independence.

2.20 Mean Independence and Covariance

- Mean independence implies
$$\mathrm{E}(y \mid x) = \mathrm{E}(y).$$
- Use the law of iterated expectations:
$$\begin{aligned}\mathrm{E}(xy) &= \mathrm{E}_x\left[\mathrm{E}(xy \mid x)\right] \\ &= \mathrm{E}_x\left[x\,\mathrm{E}(y \mid x)\right] \\ &= \mathrm{E}_x\left[x\,\mathrm{E}(y)\right] \\ &= \mathrm{E}(y)\,\mathrm{E}(x).\end{aligned}$$
- Therefore:
$$\mathrm{Cov}(x, y) = \mathrm{E}(xy) - \mathrm{E}(x)\,\mathrm{E}(y) = 0.$$
- Mean independence implies zero covariance, but not the other way around.

2.21 Annual Returns from One-Day Returns

- Thus, if daily returns are mean independent of past returns:
$$\mathrm{E}(x_{t+s} \mid x_{t+s-1}, x_{t+s-2}, \ldots, x_t) = \mathrm{E}(x_{t+s}).$$
- Then all expected product terms factor:
$$\mathrm{E}(x_t x_{t+1} x_{t+2} \cdots) = \mathrm{E}(x_t)\,\mathrm{E}(x_{t+1})\,\mathrm{E}(x_{t+2}) \cdots .$$
- If daily returns also have the same distribution,
$$\mathrm{E}(x_t) = \mathrm{E}(x_{t+1}) = \cdots = \mathrm{E}(x),$$
then the expected annual return simplifies to
$$\mathrm{E}(R_{252\mathrm{d}}) = (1 + \mathrm{E}(x))^{252} - 1.$$

2.22 Expected Annual Return

```
VT = VT %>%
  arrange(date) %>%
  mutate(price_252_days_ago = lag(price, n = 252),
         return_252d = price / price_252_days_ago - 1)

mean(VT$return_252d, na.rm = TRUE)
```

```
## [1] 0.1203372
```

```
(1 + ave_return)^252 - 1
```

```
## [1] 0.1129046
##
## These numbers are close, especially given sampling error.
##   Sampling error: variation due to randomness in the observed data.
##
## Appears consistent with mean independence across daily returns.
```

2.23 Summarizing Notions of No Dependence

- From weaker to stronger notions of no dependence.
- *Zero covariance*: no linear co-movement between y and x.

$$\text{Cov}(x, y) = \text{E}\left[(x - \text{E}(x))(y - \text{E}(y))\right] = 0.$$

- *Mean independence*: the expected value of y does not change with x.

$$\text{E}(y \mid x) = \text{E}(y).$$

- *Independence*: the distribution of y does not change with x.

$$f(y \mid x) = f(y).$$

Lecture 3

Variance

3.1 Review

- For a continuous random variable, x , the distribution can be represented by a probability density function:
$$x \sim f(x).$$
- The expected value is
$$\mathrm{E}(x) = \int x f(x)\, dx.$$
- With a sample of n observations,
$$x_1, x_2, \ldots, x_n,$$
the sample mean is
$$\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

3.2 Variance

- The mean describes the center of a distribution.
- Variance describes how spread out x is around that center.
- It is the expected squared distance from the mean:
$$\mathrm{Var}(x) = \mathrm{E}\left[(x - \mathrm{E}(x))^2\right].$$
- Define the population mean and variance:
$$\mu = \mathrm{E}(x), \qquad \sigma^2 = \mathrm{Var}(x).$$
- Then:
$$\sigma^2 = \mathrm{E}\left[(x - \mu)^2\right].$$

3.3 A Useful Formula for Variance

- Define the population mean: $\mu = \mathrm{E}(x)$.
  - μ is a fixed feature of the distribution. It is treated as a constant.
- Using properties of expectation:
$$\begin{aligned}
\mathrm{Var}(x) &= \mathrm{E}\left[(x - \mu)^2\right] \\
&= \mathrm{E}\left[x^2 - 2x\mu + \mu^2\right] \\
&= \mathrm{E}(x^2) - 2\mu\,\mathrm{E}(x) + \mu^2 \\
&= \mathrm{E}(x^2) - 2\mu^2 + \mu^2 \\
&= \mathrm{E}(x^2) - \mu^2.
\end{aligned}$$

- Therefore, variance can also be written as:

$$\text{Var}(x) = \text{E}(x^2) - [\text{E}(x)]^2.$$

3.4 Estimating Variance

- Estimating $\text{Var}(x) = \text{E}\left[(x - \text{E}(x))^2\right]$ requires an estimate of $\text{E}(x)$.
- We estimate $\text{E}(x)$ using the sample mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

- The sample variance uses deviations from the estimated center:

$$\widehat{\text{Var}(x)} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

- Using $n - 1$ instead of n is a correction (discuss later).
- In R: `var(x, na.rm = TRUE)`

3.5 Properties of Variance

- Suppose a is a constant.
- Adding a constant shifts the center but does not change the spread:

$$\text{Var}(x + a) = \text{Var}(x).$$

$$\begin{aligned}
\text{Var}(x + a) &= \text{E}\left[(x + a - \text{E}(x + a))^2\right] \\
&= \text{E}\left[(x + a - \text{E}(x) - a)^2\right] \\
&= \text{E}\left[(x - \text{E}(x))^2\right] \\
&= \text{Var}(x).
\end{aligned}$$

3.6 Properties of Variance

- Suppose a is a constant.
- Multiplying by a constant rescales the spread:

$$\text{Var}(ax) = a^2\,\text{Var}(x).$$

$$\begin{aligned}
\text{Var}(ax) &= \text{E}\left[(ax - \text{E}(ax))^2\right] \\
&= \text{E}\left[(ax - a\,\text{E}(x))^2\right] \\
&= \text{E}\left[a^2(x - \text{E}(x))^2\right] \\
&= a^2\,\text{E}\left[(x - \text{E}(x))^2\right] \\
&= a^2\,\text{Var}(x).
\end{aligned}$$

3.7 Variance of Sum of Two Random Variables

- Expectation is linear:
$$\mathrm{E}(x + y) = \mathrm{E}(x) + \mathrm{E}(y).$$
- Variance is different:
$$\mathrm{Var}(x + y) = \mathrm{Var}(x) + \mathrm{Var}(y) + 2\,\mathrm{Cov}(x, y).$$
- Let
$$\tilde{x} = x - \mathrm{E}(x), \qquad \tilde{y} = y - \mathrm{E}(y).$$

$$\begin{aligned}
\mathrm{Var}(x + y) &= \mathrm{E}\left[(x + y - \mathrm{E}(x + y))^2\right] \\
&= \mathrm{E}\left[(\tilde{x} + \tilde{y})^2\right] \\
&= \mathrm{E}\left[\tilde{x}^2 + 2\tilde{x}\tilde{y} + \tilde{y}^2\right] \\
&= \mathrm{E}(\tilde{x}^2) + 2\,\mathrm{E}(\tilde{x}\tilde{y}) + \mathrm{E}(\tilde{y}^2) \\
&= \mathrm{Var}(x) + 2\,\mathrm{Cov}(x, y) + \mathrm{Var}(y).
\end{aligned}$$

3.8 Linear Function to Two Random Variables

- If x and y are random variables and a and b are constants, then
$$\mathrm{E}(ax + by) = a\,\mathrm{E}(x) + b\,\mathrm{E}(y).$$

$$\mathrm{Var}(ax + by) = a^2\,\mathrm{Var}(x) + b^2\,\mathrm{Var}(y) + 2ab\,\mathrm{Cov}(x, y).$$

3.9 Standard Deviation

- Variance is measured in squared units.
  - If x is measured in dollars, $\mathrm{Var}(x)$ is measured in dollars squared.
- The standard deviation puts spread back in the original units:
$$\mathrm{SD}(x) = \sqrt{\mathrm{Var}(x)}, \qquad \sigma = \sqrt{\sigma^2}.$$
- The sample standard deviation is
$$\widehat{\mathrm{SD}(x)} = \sqrt{\widehat{\mathrm{Var}(x)}}.$$
- In R:
`sd(x, na.rm = TRUE)`, or `sqrt(var(x, na.rm = TRUE))`
- For roughly normal shaped distributions, most values lie within about $\pm 2\sigma$ of the mean.

3.10 Variance of Returns

- VTI is a total U.S. stock market index fund.
  - Broad exposure to U.S. stocks.
- AGG is a total U.S. bond market index fund.
  - Broad exposure to investment-grade U.S. bonds.
- Variance of returns measures how unpredictable returns are.
  - In finance, this is one way to measure *risk*.
- How do stocks and bonds compare in average return and risk?

3.11 Yearly Returns

```
getSymbols(c('VTI','AGG'),
           from = "2004-01-01",
           to = floor_date(Sys.Date(),'year') - 1)
```

```
## [1] "VTI" "AGG"
```

```
return_data = merge(VTI %>% Ad() %>% yearlyReturn(),
                    AGG %>% Ad() %>% yearlyReturn()) %>%
  as.data.frame()
names(return_data) = c('VTI', 'AGG')

tail(return_data, 4)
```

```
##                    VTI         AGG
## 2022-12-30 -0.1952344 -0.13021616
## 2023-12-29  0.2604697  0.05655262
## 2024-12-31  0.2381024  0.01309262
## 2025-12-30  0.1799935  0.07449358
```

```
mean(return_data$VTI)
var(return_data$VTI)
sd(return_data$VTI)
```

3.12 Yearly Returns

```
## There are 22 years of annual returns
##
## ------------------ VTI ------------------
##
## For VTI, the average annual return is 0.12171
##
## The variance of annual returns is 0.02814
## The standard deviation of annual returns is 0.16775
##
## ------------------ AGG ------------------
##
## For AGG, the average annual return is 0.03281
##
## The variance of annual returns is 0.00228
## The standard deviation of annual returns is 0.04777
```

3.13 Risk-Adjusted Return

- VTI has a higher average monthly return.
- AGG has a lower average monthly return, but much lower risk.
- A simple return-to-risk ratio is:

$$\frac{\mathrm{E}(r)}{\mathrm{SD}(r)}.$$

- This measures average return per unit of volatility.
- A higher value means more average return for each unit of risk.
- What is the ratio for VTI and AGG?

3.14 Portfolio Returns

- Suppose a portfolio invests in VTI and AGG.
- Let VTI_t and AGG_t be their returns in period t.
- An 80/20 portfolio puts 80% in VTI and 20% in AGG:

$$R_t = 0.8\mathrm{VTI}_t + 0.2\mathrm{AGG}_t.$$

- The expected portfolio return is:

$$\mathrm{E}(R_t) = 0.8\,\mathrm{E}(\mathrm{VTI}_t) + 0.2\,\mathrm{E}(\mathrm{AGG}_t).$$

- The variance of the portfolio return is:

$$\begin{aligned}\mathrm{Var}(R_t) = {} & 0.8^2\,\mathrm{Var}(\mathrm{VTI}_t) + 0.2^2\,\mathrm{Var}(\mathrm{AGG}_t) \\ & + 2(0.8)(0.2)\,\mathrm{Cov}(\mathrm{VTI}_t, \mathrm{AGG}_t).\end{aligned}$$

3.15 Portfolio Returns

```
return_data = return_data %>%
  mutate(portfolio = 0.8*VTI + 0.2*AGG)
tail(return_data, 3)
```

```
##                   VTI        AGG portfolio
## 2023-12-29 0.2604697 0.05655262 0.2196863
## 2024-12-31 0.2381024 0.01309262 0.1931004
## 2025-12-30 0.1799935 0.07449358 0.1588935
```

```
cov(return_data$VTI, return_data$AGG)
```

```
## [1] 0.001494462
```

```
(0.8^2)*var(return_data$VTI) + (0.2^2)*var(return_data$AGG) +
  2*0.8*0.2*cov(return_data$VTI, return_data$AGG)
```

```
## [1] 0.01857938
##
##  The average annual return of portfolio is 0.10393
##
##  The variance of annual returns of portfolio is 0.01858
##  The standard deviation of annual returns of portfolio is 0.13631
```

3.16 Linear Functions of Random Variables

- A linear function of many random variables can produce many covariance terms.
- If $x_1, \ldots, x_n$ are random variables and $a_1, \ldots, a_n$ are constants, then

$$\operatorname{Var}\left(\sum_{i=1}^{n} a_i x_i\right) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(x_i) + 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} a_i a_j \operatorname{Cov}(x_i, x_j).$$

- This simplifies only if the variables are uncorrelated:

$$\operatorname{Var}\left(\sum_{i=1}^{n} a_i x_i\right) = \sum_{i=1}^{n} a_i^2 \operatorname{Var}(x_i).$$

- Matrix notation gives a compact way to keep track of all the covariance terms.

3.17 Variance of a Vector Product

- Let x be a column vector of random variables and a be a column vector of constants:

$$x = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}', \qquad a = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}'.$$

- The scalar $a'x$ is a weighted sum:

$$a'x = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n.$$

- This gives the sandwich form for the variance

$$\operatorname{Var}\left(\sum_{i=1}^{n} a_i x_i\right) = \operatorname{Var}(a'x) = a' \operatorname{Var}(x) a.$$

- $\operatorname{Var}(x)$ is an $n \times n$ variance-covariance matrix.

3.18 Variance-Covariance Matrix

- For a vector of random variables,

$$x = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}',$$

  the variance of x is

$$\operatorname{Var}(x) = \begin{bmatrix} \operatorname{Var}(x_1) & \operatorname{Cov}(x_1, x_2) & \cdots & \operatorname{Cov}(x_1, x_n) \\ \operatorname{Cov}(x_2, x_1) & \operatorname{Var}(x_2) & \cdots & \operatorname{Cov}(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \operatorname{Cov}(x_n, x_1) & \operatorname{Cov}(x_n, x_2) & \cdots & \operatorname{Var}(x_n) \end{bmatrix}.$$

- The diagonal entries are variances.
- The off-diagonal entries are covariances.

3.19 Portfolio Returns in Vector Form

- Stack the asset returns in a vector:

$$x_t = \begin{bmatrix} \mathrm{VTI}_t \\ \mathrm{AGG}_t \end{bmatrix}.$$

- Stack the portfolio weights in a vector:

$$a = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}.$$

- Portfolio return is a weighted sum:

$$R_t = a' x_t.$$

- Written out:

$$a' x_t = \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \begin{bmatrix} \mathrm{VTI}_t \\ \mathrm{AGG}_t \end{bmatrix} = 0.8\mathrm{VTI}_t + 0.2\mathrm{AGG}_t.$$

- The vector product produces one number: the portfolio return.

3.20 Variance of Portfolio Return

- The return vector contains two random variables:

$$x_t = \begin{bmatrix} \text{VTI}_t \\ \text{AGG}_t \end{bmatrix}.$$

- Its variance is a variance-covariance matrix:

$$\begin{aligned} \text{Var}(x_t) &= \begin{bmatrix} \text{Var}(\text{VTI}_t) & \text{Cov}(\text{VTI}_t, \text{AGG}_t) \\ \text{Cov}(\text{AGG}_t, \text{VTI}_t) & \text{Var}(\text{AGG}_t) \end{bmatrix} \\ &= \begin{bmatrix} \sigma^2_{\text{VTI}} & \sigma_{\text{VTI,AGG}} \\ \sigma_{\text{AGG,VTI}} & \sigma^2_{\text{AGG}} \end{bmatrix}. \end{aligned}$$

- Since $R_t = a' x_t$, portfolio variance is

$$\text{Var}(R_t) = a' \, \text{Var}(x_t) a.$$

3.21 Expanding Portfolio Variance

- For the 80/20 portfolio:

$$\begin{aligned} \text{Var}(R_t) &= \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \text{Var}(x_t) \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \begin{bmatrix} \sigma^2_{\text{VTI}} & \sigma_{\text{VTI,AGG}} \\ \sigma_{\text{VTI,AGG}} & \sigma^2_{\text{AGG}} \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \begin{bmatrix} 0.8\sigma^2_{\text{VTI}} + 0.2\sigma_{\text{VTI,AGG}} \\ 0.8\sigma_{\text{VTI,AGG}} + 0.2\sigma^2_{\text{AGG}} \end{bmatrix} \\ &= 0.8(0.8\sigma^2_{\text{VTI}} + 0.2\sigma_{\text{VTI,AGG}}) \\ &\quad + 0.2(0.8\sigma_{\text{VTI,AGG}} + 0.2\sigma^2_{\text{AGG}}) \\ &= 0.8^2\sigma^2_{\text{VTI}} + 0.2^2\sigma^2_{\text{AGG}} + 2(0.8)(0.2)\sigma_{\text{VTI,AGG}}. \end{aligned}$$

- This is the same variance-of-a-sum formula, written in matrix form.

3.22 Variance of Portfolio Returns Using Matrix

```
Vx = cov(return_data[,c('VTI', 'AGG')])
Vx
```

```
##             VTI         AGG
## VTI 0.028140453 0.001494462
## AGG 0.001494462 0.002281511
```

```
a = rbind(0.8, 0.2)

t(a) %*% Vx %*% a
```

```
##             [,1]
## [1,] 0.01857938
```

Lecture 4

Estimation

4.1 Estimation

- Data come from an underlying process.
- Knowing this process would help guide decisions.
- We typically summarize the process with a finite number of unknown *parameters*.
- Since the process generated the data, we can use the data to learn about the parameters.
  - This is the idea behind *parameter estimation*.
- How much can we learn about the parameter from the data?
- How close is our estimate to the actual parameter?

4.2 Estimation Example

- Suppose we have 36 months of past stock returns.
- We might ask: what will the return be next month?
- The exact return next month is random, so we cannot know it in advance.
- But knowing the distribution of returns would help:

$$x_{t+1} \sim f.$$

- For example, knowing $\mathrm{E}(x)$ would tell us the expected return.

4.3 Estimation Example

- Given our sample of 36 returns, a reasonable estimator for $\mathrm{E}(x)$ is the sample mean:

$$\bar{x} = \frac{1}{36} \sum_{i=1}^{36} x_i.$$

- $\bar{x}$ will be different from $\mathrm{E}(x)$, but hopefully it is close.
- *Key Point:* The data are random, so the estimator is random:

$$x_1, \ldots, x_{36} \text{ random} \qquad \Rightarrow \qquad \bar{x} \text{ random}.$$

- Therefore, $\bar{x}$ has its own distribution.
  - This is the *sampling distribution* of the estimator.
- We evaluate how well $\bar{x}$ recovers $\mathrm{E}(x)$ by studying this distribution.

4.4 Sampling Properties of Estimators

- Let $\hat{\mu}$ be an estimator of an unknown parameter μ.
- $\hat{\mu}$ is a random variable because computed from random data.
- We evaluate $\hat{\mu}$ by studying its distribution.
- Desirable properties include:
  - *Unbiasedness:* $\mathrm{E}(\hat{\mu}) = \mu$
  - *Consistency:* $\hat{\mu} \to \mu$ as $n \to \infty$
  - *Efficiency:* small variance relative to other estimators
  - *Approximate normality:* useful for inference

4.5 Using an Estimator

- Once we have an estimator and know its sampling properties, what can we do?
- Point estimation: $\hat{\mu}$ is a single "best guess" of the unknown parameter.
- Interval estimation: construct a range of plausible values for μ.
  - Constructed from data → the interval itself is random.
  - Sampling properties tell us how often this procedure works.
- Hypothesis testing: formally assess claims about the underlying parameter.

4.6 Is the Sample Average a Good Estimator?

- Suppose x_i is a random variable with

$$\mathrm{E}(x_i) = \mu, \qquad \mathrm{Var}(x_i) = \sigma^2.$$

- We observe a random sample of n observations:

$$x_1, x_2, \ldots, x_n.$$

  - Each observation is independent and has the same distribution (i.i.d).
- Common estimators for μ and σ^2 are their sample analogs:

$$\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

- Based on analogy principle: replace population quantities with sample counterparts.

4.7 Deriving the Sampling Properties

- The estimator is
$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$
- Summation notation is cumbersome. Rewrite as
$$\hat{\mu} = a'x.$$
- Where
$$a = \begin{bmatrix} 1/n \\ 1/n \\ \vdots \\ 1/n \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$
- a is a vector of constants. x is a vector of random variables.

4.8 The Joint Distribution of the Data

- Each x_i is i.i.d.: independent and identically distributed.
- Since each x_i comes from the same distribution, each has the same mean:
$$\mathrm{E}(x) = \begin{bmatrix} \mathrm{E}(x_1) \\ \mathrm{E}(x_2) \\ \vdots \\ \mathrm{E}(x_n) \end{bmatrix} = \begin{bmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{bmatrix}.$$

4.9 The Joint Distribution of the Data

- The variance-covariance matrix of the data is
$$\mathrm{Var}(x) = \begin{bmatrix} \mathrm{Var}(x_1) & \mathrm{Cov}(x_1, x_2) & \cdots & \mathrm{Cov}(x_1, x_n) \\ \mathrm{Cov}(x_2, x_1) & \mathrm{Var}(x_2) & \cdots & \mathrm{Cov}(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(x_n, x_1) & \mathrm{Cov}(x_n, x_2) & \cdots & \mathrm{Var}(x_n) \end{bmatrix}.$$
- Each x_i is independent, so covariances are zero.
- Each x_i has the same distribution, so each has the same variance:
$$\mathrm{Var}(x) = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_n.$$

4.10 Expected Value of the Sample Mean

- x is a random vector so $\hat{\mu} = a'x$ is a random variable.

$$\begin{aligned}
\mathrm{E}(\hat{\mu}) &= \mathrm{E}(a'x) \\
&= a' \, \mathrm{E}(x) \\
&= \begin{bmatrix} 1/n & 1/n & \cdots & 1/n \end{bmatrix} \begin{bmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{bmatrix}_{n \times 1} \\
&= \frac{1}{n}\mu + \frac{1}{n}\mu + \cdots + \frac{1}{n}\mu \\
&= \mu.
\end{aligned}$$

- $\hat{\mu}$ is *unbiased*: its expected value equals the parameter we are trying to estimate.

4.11 Variance of the Sample Mean

$$\begin{aligned}
\mathrm{Var}(\hat{\mu}) &= \mathrm{Var}(a'x) \\
&= a' \, \mathrm{Var}(x) a \\
&= a'(\sigma^2 I_n) a \\
&= \sigma^2 a' a \\
&= \sigma^2 \begin{bmatrix} 1/n & \cdots & 1/n \end{bmatrix} \begin{bmatrix} 1/n \\ \vdots \\ 1/n \end{bmatrix} \\
&= \sigma^2 \left(\frac{1}{n^2} + \frac{1}{n^2} + \cdots + \frac{1}{n^2} \right) \\
&= \frac{\sigma^2}{n}.
\end{aligned}$$

- As n gets larger, $\mathrm{Var}(\hat{\mu})$ gets smaller.

4.12 Consistency of the Sample Mean

- The sample mean is unbiased:

$$\mathrm{E}(\hat{\mu}) = \mu.$$

- Its variance shrinks with sample size:

$$\mathrm{Var}(\hat{\mu}) = \frac{\sigma^2}{n}.$$

- As $n \to \infty$,

$$\mathrm{Var}(\hat{\mu}) \to 0.$$

- Therefore, the sampling distribution of $\hat{\mu}$ collapses around μ.
- This is the intuition for consistency:

$$\hat{\mu} \to \mu.$$

4.13 The Sample Mean is Normally Distributed

- Two ways to show $\hat{\mu}$ is normally distributed:
- Assume normality of the data:
  - If each $x_i \sim N(\mu, \sigma^2)$, then $\hat{\mu}$ is a sum of normal random variables.
  - Sums of normal random variables are also normal.
- Use the *Central Limit Theorem*:
  - With a large sample, the sample average is approximately normal.
  - This holds even when the underlying data are not normally distributed.

4.14 The Sampling Distribution of $\hat{\mu}$

- Putting it all together:

$$\hat{\mu} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

- Standardizing the estimator:

$$\frac{\hat{\mu} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1).$$

- Main challenge: we do not know σ^2.
  - We estimate it using

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

  - Using $\hat{\sigma}^2$ in place of σ^2 changes the sampling distribution.

4.15 Understanding the Variance of the Estimator

- The sample variance estimates $\text{Var}(x_i) = \sigma^2$:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

- Since

$$\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n},$$

an estimate of the variance of $\hat{\mu}$ is

$$\widehat{\text{Var}(\hat{\mu})} = \frac{\hat{\sigma}^2}{n}.$$

- The estimated standard error is

$$\widehat{\mathrm{SE}(\hat{\mu})} = \sqrt{\widehat{\mathrm{Var}(\hat{\mu})}} = \sqrt{\frac{\hat{\sigma}^2}{n}}.$$

4.16 Working with the Distribution

- If σ^2 were known, then

$$\frac{\hat{\mu} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1).$$

- But σ^2 is unknown, so we use the estimated standard error:

$$\frac{\hat{\mu} - \mu}{\widehat{\mathrm{SE}(\hat{\mu})}}.$$

- The denominator is now also random.
- This changes the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\mathrm{SE}(\hat{\mu})}} \sim t_{n-1}.$$

4.17 Working with the Distribution

- We have $\hat{\mu}$, consistent and unbiased point estimate.
- We also know the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\mathrm{SE}(\hat{\mu})}} \sim t_{n-1}.$$

- This expression has four components:
  - the estimate, $\hat{\mu}$
  - the unknown parameter, μ
  - the estimated standard error, $\widehat{\mathrm{SE}(\hat{\mu})}$
  - the t_{n-1} distribution
- We know three of them. We invert the expression to learn about the unknown component: μ.

4.18 Estimating the Mean Monthly Returns

```
VTI = getSymbols('VTI',
                 from = floor_date(Sys.Date(),'month') - years(3),
                 to = floor_date(Sys.Date(),'month') - 1,
                 auto.assign = FALSE) %>%
  Ad() %>%
  monthlyReturn() %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date))

tail(VTI, 5)
```

```
##          date monthly.returns
## 32 2026-03-31    -0.050048376
## 33 2026-04-30     0.104017878
## 34 2026-05-29     0.051838149
## 35 2026-06-30    -0.003853479
## 36 2026-07-30    -0.010188140
```

```
nrow(VTI)
```

```
## [1] 36
```

4.19 Estimating the Mean Monthly Returns

```
n = nrow(VTI)

ave_return = mean(VTI$monthly.returns)
var_return = var(VTI$monthly.returns)
sd_return = sd(VTI$monthly.returns)

mu_hat = ave_return
mu_hat_var = var_return/n
mu_hat_se = sqrt(mu_hat_var)
```

```
##
##  The average monthly return is 0.01512
##  The standard deviation of monthly returns is 0.03891
##  The standard error of the estimate is 0.00649
```

Lecture 5

Inference

5.1 Reviewing Estimation

- From our sample, we calculate an estimate of an unknown parameter μ.
- Our point estimate $\hat{\mu}$ is a random variable.
- If $\hat{\mu}$ is unbiased and normally distributed, then

$$\hat{\mu} \sim N\left(\mu, \text{Var}(\hat{\mu})\right).$$

- We estimate the variance of the estimator using assumptions and data:

$$\widehat{\text{Var}(\hat{\mu})}, \qquad \widehat{\text{SE}(\hat{\mu})} = \sqrt{\widehat{\text{Var}(\hat{\mu})}}.$$

- The finite-sample distribution we use is

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}(\hat{\mu})}} \sim t_{n-1}.$$

5.2 Inference

We can use the sampling distribution to:

1) Create interval estimates, or confidence intervals.
   - A range of plausible values for μ.
2) Conduct hypothesis tests.
   - A formal way to assess claims about a particular value of μ.

5.3 Interval Estimation

- Our estimate may be high or low simply because of sampling randomness.
  - We do not know whether our estimate landed in the lower tail, center, or upper tail.
- But we do know the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\text{SE}(\hat{\mu})}} \sim t_{n-1}.$$

- Most estimates will be within about two standard errors of the true parameter.
- Equivalently, the true parameter will usually be within about two standard errors of the estimate.
- This idea leads to a *confidence interval*.

5.4 Interval Estimation

- $\hat{\mu}$ is our best guess of μ.
  - But $\hat{\mu}$ will generally not exactly equal μ.
- Interval estimation uses the data to identify a range of plausible values:

$$[\text{CI}_{\text{lower}}, \text{CI}_{\text{upper}}].$$

- The confidence level c means:

$$\Pr(\text{CI}_{\text{lower}} \leq \mu \leq \text{CI}_{\text{upper}}) = c.$$

- Interpretation: $(1 - c) \times 100\%$ of the time, the interval fails to cover μ.
  - It is not wide enough to reach μ on one side or the other.

5.5 Interval Estimation

- Interval estimation begins with the sampling distribution.
  - In this case, the t_{n-1} distribution.
- For a given confidence level c, we can find values $\underline{t}$ and $\overline{t}$ such that

$$\Pr\left(\underline{t} \leq t \leq \overline{t}\right) = c.$$

- Since the t distribution is symmetric around zero, define one critical value:

$$\kappa = \overline{t} = -\underline{t}.$$

- The expression we work with is

$$\Pr\left(-\kappa \leq t \leq \kappa\right) = c.$$

5.6 Finding the Critical Value

- Given the t_{n-1} distribution, for any confidence level c we can find κ such that

$$\Pr\left(-\kappa \leq t \leq \kappa\right) = c.$$

- The uncovered probability is split equally across the two tails:

$$\Pr(t > \kappa) = \frac{1 - c}{2}, \qquad \Pr(t < -\kappa) = \frac{1 - c}{2}.$$

- We calculate κ using the inverse CDF:

`crit_value = -qt((1 - conf_lvl)/2, df=n-1)`.

5.7 Interval Estimation Math

- We start with the sampling distribution:

$$\frac{\hat{\mu} - \mu}{\widehat{\mathrm{SE}(\hat{\mu})}} \sim t_{n-1}.$$

- Given confidence level c, choose κ such that

$$\Pr\left(-\kappa \leq \frac{\hat{\mu} - \mu}{\widehat{\mathrm{SE}(\hat{\mu})}} \leq \kappa\right) = c.$$

- Rearranging gives:

$$\Pr\left(\hat{\mu} - \kappa\widehat{\mathrm{SE}(\hat{\mu})} \leq \mu \leq \hat{\mu} + \kappa\widehat{\mathrm{SE}(\hat{\mu})}\right) = c.$$

- Therefore, the confidence interval is

$$\left[\hat{\mu} - \kappa\widehat{\mathrm{SE}(\hat{\mu})}, \quad \hat{\mu} + \kappa\widehat{\mathrm{SE}(\hat{\mu})}\right].$$

5.8 Steps to Interval Estimation

(1) Choose confidence level c.

  - Example: $c = 0.95$, a 95% confidence interval.

(2) Choose critical value κ from the t_{n-1} distribution:

$$\Pr(t > \kappa) = \frac{1 - c}{2}.$$

(3) Construct the interval:

$$\left[\hat{\mu} - \kappa\widehat{\mathrm{SE}(\hat{\mu})}, \quad \hat{\mu} + \kappa\widehat{\mathrm{SE}(\hat{\mu})}\right].$$

- Tradeoff:
  - Higher confidence $\Rightarrow$ larger κ $\Rightarrow$ wider interval.
  - Wider intervals are more likely to cover μ, but less precise.

5.9 Confidence Interval for Mean Monthly Return

```
VTI = getSymbols('VTI',
                 from = floor_date(Sys.Date(),'month') - years(3),
                 to = floor_date(Sys.Date(),'month') - 1,
                 auto.assign = FALSE) %>%
  Ad() %>%
  monthlyReturn() %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date))
n = nrow(VTI)
mu_hat = mean(VTI$monthly.returns)
mu_hat_se = sqrt(var(VTI$monthly.returns)/n)
```

```
##
## The estimated expected monthly return is 0.01512
##
## The standard error of the estimate is 0.00649
```

5.10 Confidence Interval for Mean Monthly Return

```
crit_val_95 = -qt((1 - 0.95)/2, df = n - 1)
mu_hat_lb_95 = mu_hat - crit_val_95*mu_hat_se
mu_hat_ub_95 = mu_hat + crit_val_95*mu_hat_se

crit_val_99 = -qt((1 - 0.99)/2, df = n - 1)
mu_hat_lb_99 = mu_hat - crit_val_99*mu_hat_se
mu_hat_ub_99 = mu_hat + crit_val_99*mu_hat_se
```

```
##
## The critical value for 95% CI is 2.03011
##
## The lower bound 95% CI is 0.00196
##
## The upper bound 95% CI is 0.02829
##
## The critical value for 99% CI is 2.72381
##
## The lower bound 99% CI is -0.00254
##
## The upper bound 99% CI is 0.03279
```

5.11 Hypothesis Testing

- Consider the *null hypothesis*: $\mu = \mu_0$.
- If the null is true, the sampling distribution of $\hat{\mu}$ is

$$\hat{\mu} \sim N\left(\mu_0, \text{Var}(\hat{\mu})\right).$$

- We expect most values of $\hat{\mu}$ to be close to μ_0.

- If $\hat{\mu}$ is far from μ_0, this could happen because:

  1) an unusual sample produced an unusual estimate.
  2) the null hypothesis is wrong.

- Hypothesis testing formalizes when (1) becomes unlikely enough to reject the null.

5.12 Hypothesis Testing of a Single Parameter

- In a two-sided test, we state the null hypothesis:

$$H_0 : \mu = \mu_0.$$

- The alternative hypothesis is:

$$H_1 : \mu \neq \mu_0.$$

- Hypothesis testing asks:
  - If the null is true, how likely are we to observe our estimate, or one more extreme?
  - This probability is called the *p-value*.
- If the p-value is below a chosen threshold, such as 0.05 or 0.01, we reject the null.
  - *Rejecting the null* means the data provide evidence that $\mu \neq \mu_0$.

5.13 T-Test

- From the sample, we estimate $\hat{\mu}$ and $\widehat{\mathrm{SE}(\hat{\mu})}$.
- Consider the hypotheses:

$$H_0 : \mu = \mu_0, \qquad H_1 : \mu \neq \mu_0.$$

- Calculate the t-statistic:

$$\mathrm{t} = \frac{\hat{\mu} - \mu_0}{\widehat{\mathrm{SE}(\hat{\mu})}}.$$

  - The t-statistic measures how far $\hat{\mu}$ is from μ_0, adjusted for sampling uncertainty.
  - If the null is true, this statistic follows a t_{n-1} distribution, centered at zero.
  - Large absolute values of t are evidence against the null.

5.14 P-Value

- For every hypothesis test, we can calculate a *p-value*.
  - If the null is true, this is the probability of observing a test statistic at least as extreme as ours.
- Let $F_{t_{n-1}}(a)$ denote the CDF of a t distribution with $n-1$ degrees of freedom.

- Since the t-statistic follows this distribution under the null, for a two-sided test:

$$\text{p-value} = 2 \times \left[1 - F_{t_{n-1}}\left(|\text{t}|\right)\right].$$

- In R:

```
2 * (1 - pt(abs(tstat), df = n - 1))
```

5.15 Significance Level

- To conclude the hypothesis test, we choose a significance level, α.
- α is the probability of rejecting the null when the null is true.
  - $\alpha = 0.05$: 1/20 times reject a true null.
  - $\alpha = 0.01$: 1/100 times reject a true null.
- Decision rule:
  - If p-value $< \alpha$, *reject the null* at significance level α.
    - The data provide evidence against the null.
  - If p-value $\geq \alpha$, *fail to reject the null*.

5.16 Test Mean Monthly Return VTI is 0

```
mu_null = 0
t_stat = (mu_hat - mu_null)/mu_hat_se
p_val = 2*(1 - pt(abs(t_stat), df = n - 1))
```

```
##
## The t-statistic is 2.33169
##
## The p-value is 0.02560
##
## With our 36 observations
##
## Reject null that the month return to VTI is zero at the 5% level.
##
## However, fail to reject at the 1% level.
```

5.17 Failing to Reject the Null

- We fail to reject the null when t is close to zero.:

$$\text{t} = \frac{\hat{\mu} - \mu_0}{\widehat{\text{SE}(\hat{\mu})}}.$$

- This can happen for two reasons:

  1) $\hat{\mu}$ is close to μ_0.
  2) $\widehat{\text{SE}(\hat{\mu})}$ is large, so the estimate is imprecise.

- Failing to reject should be interpreted cautiously.
  - If $\widehat{\text{SE}(\hat{\mu})}$ is small, failing to reject may provide some support for the null.
  - If $\widehat{\text{SE}(\hat{\mu})}$ is large, the test may simply be inconclusive.

5.18 Critical Value

- We can conduct a hypothesis test at significance level α using a critical value.
- For a two-sided test with test statistic from a t_{n-1} distribution, choose κ such that

$$\Pr(t > \kappa) = \frac{\alpha}{2}.$$

- The rejection region is

$$t < -\kappa \quad \text{or} \quad t > \kappa$$

- Decision rule:
  - Reject the null if $|\text{t}| > \kappa$.
  - Fail to reject the null if $|\text{t}| \leq \kappa$.
- In R: `crit_value = -qt(alpha / 2, df = n - 1)`

5.19 Hypothesis Tests and Confidence Intervals

- The critical value for a two-sided test at significance level α is

`crit_value = -qt(alpha / 2, df = n - 1)`.

- This is the same critical value used for a $(1 - \alpha)$ confidence interval.
- Therefore, we can use the confidence interval to conduct the hypothesis test.
- Example: given a 95% confidence interval,
  - If μ_0 is inside the confidence interval, fail to reject $H_0 : \mu = \mu_0$ at the 5% level.
  - If μ_0 is outside the confidence interval, reject $H_0 : \mu = \mu_0$ at the 5% level.

Lecture 6

Simulation Methods

6.1 Simulation Methods

- Many statistical questions are easy to ask but hard to answer analytically. The algebra is difficult.
  - Example: what is the sampling distribution of a nonlinear estimator?
- The challenge is that we only observe only one random sample.
  - If we observed many random samples, the question would be easy to answer.
- Simulation methods create artificial datasets that can be used to answer a broad range of questions.
  - These methods play a growing role with advances in modern computing power.

6.2 Correlation

- Correlation is standardized covariance:

$$\rho = \text{Cor}(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x)\,\text{Var}(y)}}.$$

- Measures linear co-movement in SD units.
  - It does not depend on the units of x or y.
- If $\text{Cor}(x, y) = 1$, one SD higher in x corresponds to one SD higher in y.
- Correlation takes values between -1 and 1.
  - 1: perfect positive linear relationship
  - 0: no linear relationship
  - -1: perfect negative linear relationship

6.3 Estimating Correlation

- Correlations are straightforward to estimate from the sample:

$$\hat{\rho} = \widehat{\text{Cor}(x, y)} = \frac{\widehat{\text{Cov}(x, y)}}{\sqrt{\widehat{\text{Var}(x)}\widehat{\text{Var}(y)}}}.$$

- But deriving the standard error,

$$\text{SE}\left(\widehat{\text{Cor}(x, y)}\right),$$

  is less straightforward.
- The statistic depends on a complicated nonlinear function of the sample.
- This type of problem is a natural candidate for bootstrapping.

6.4 Returns to Stocks and Bonds

```
getSymbols(c('VTI', 'AGG'),
           from = "2004-01-01",
           to = floor_date(Sys.Date(),'month') - 1)
```

```
## [1] "VTI" "AGG"
```

```
VTI = VTI %>%
  Ad() %>%
  monthlyReturn()
names(VTI) = 'VTI'

AGG = AGG %>%
  Ad() %>%
  monthlyReturn()
names(AGG) = 'AGG'

ETF_data = merge(VTI, AGG) %>%
  as.data.frame() %>%
  tibble::rownames_to_column('date') %>%
  mutate(date = as_date(date))
```

```
##           date          VTI          AGG
## 269 2026-05-29  0.051838149  0.002936027
## 270 2026-06-30 -0.003853479  0.002542450
## 271 2026-07-30 -0.010188140 -0.010430929
##
## Data has 271 obeservations of monthly returns
```

6.5 Correlation of Returns of Stocks and Bonds

```
cor(ETF_data$VTI, ETF_data$AGG)
```

```
## [1] 0.2298297
```

```
cov(ETF_data$VTI, ETF_data$AGG)/
  sqrt(var(ETF_data$VTI)*var(ETF_data$AGG))
```

```
## [1] 0.2298297
```

```
cov(ETF_data$VTI, ETF_data$AGG)/
  (sd(ETF_data$VTI)*sd(ETF_data$AGG))
```

```
## [1] 0.2298297
```

6.6 Bootstrapping a Standard Error

- $\widehat{\text{SE}(\hat{\rho})}$ measures how much $\hat{\rho}$ varies due to random sampling.
- If we had B samples, we could calculate B estimates:

$$\hat{\rho}^*_{(1)}, \hat{\rho}^*_{(2)}, \ldots, \hat{\rho}^*_{(B)}.$$

- Then we could estimate the standard error directly:

$$\widehat{\text{SE}(\hat{\rho})} = \sqrt{\frac{1}{B-1}\sum_{s=1}^{B}\left(\hat{\rho}^*_{(s)} - \bar{\hat{\rho}}^*\right)^2}.$$

- Bootstrapping calculates the standard error by creating artificial samples by resampling from the original data

6.7 Comments on Bootstrapping

- Bootstrap logic: our data are a random sample from the population.
  - Resampling from the data approximates sampling from the population.
- Bootstrap standard errors can differ from analytical formulas.
  - They may be more credible when analytical assumptions are questionable.
- There are many bootstrap strategies.
  - Non-parametric bootstrap, parametric bootstrap, wild bootstrap, block bootstrap, etc.

6.8 Non-Parametric Bootstrapping

- We have n pairs of annual returns for VTI and AGG:

$$(\text{VTI}_1, \text{AGG}_1), (\text{VTI}_2, \text{AGG}_2), \ldots, (\text{VTI}_n, \text{AGG}_n).$$

1) Create B samples by sampling n pairs *with replacement*.
   - Larger B gives a more accurate bootstrap approximation.
2) For each sample, calculate $\hat{\rho}^*_{(s)}$.
3) Estimate the standard error:

$$\widehat{\text{SE}(\hat{\rho})} = \text{SD}\left(\hat{\rho}^*_{(s)}\right).$$

6.9 Bootstrapping a Standard Error: Expanding

```
rho_hat = cor(ETF_data$VTI, ETF_data$AGG)

n_boots = 5000
n_obs = nrow(ETF_data)
boot_data = ETF_data %>%
  slice_sample(n = n_boots*n_obs, replace = TRUE) %>%
  mutate(boot_idx = rep(1:n_boots, each = n_obs)) %>%
  group_by(boot_idx) %>%
  summarize(rho_hat = cor(VTI, AGG))

rho_hat_se = sd(boot_data$rho_hat)
```

```
##
## The standard error of the estimate is 0.07045
```

6.10 Bootstrapping a Standard Error: Loop

```
n_boots = 5000
n_obs = nrow(ETF_data)
boot_data = vector(mode = 'numeric', length = n_boots)
for (s in 1:n_boots){
  samp_idx = sample(n_obs, n_obs, replace = TRUE)
  boot_data[s] = cor(ETF_data$VTI[samp_idx], ETF_data$AGG[samp_idx])
}
rho_hat_se = sd(boot_data)
```

```
##
## The standard error of the estimate is 0.07011
```

6.11 Correlation of Returns of Stocks and Bonds

```
rho_null = 0
tstat = (rho_hat - rho_null)/rho_hat_se
pval = 2*(1 - pt(abs(tstat), df = n_obs - 1))

crit_val = -qt((1-0.95)/2, df = n_obs - 1)

CI95 = rho_hat + c(-1, 1)*crit_val*rho_hat_se
```

```
##
## The number of observations is 271.
##
## The estimated correlation is 0.22983.
##
## The standard error of the estimate is 0.07011.
##
## Testing H0: rho = 0.00, the t-statistic is 3.278.
##
## The p-value for the test is 0.0012.
##
## The critical value for a 95% CI is 1.969.
##
## The 95% confidence interval is [0.09180, 0.36786].
```

6.12 Monte Carlo Simulation

- We have about 20 years of return data.
- Suppose we want to study ten-year returns.
  - For example: mean, variance, probability of losing money.
- With only 20 years of data, we have only about two non-overlapping ten-year returns.
  - Not enough to estimate these quantities reliably.
- Overlapping ten-year returns create more observations.
  - But the observations are highly dependent, which creates compications

6.13 Monte Carlo Simulation

- If we had a sample of ten-year returns,

$$R_{10\text{Y},1}, R_{10\text{Y},2}, R_{10\text{Y},3}, \ldots, R_{10\text{Y},n},$$

  estimating the mean and variance would be straightforward.
- Monte Carlo simulation uses the data to create many artificial ten-year return histories.
- Treat the simulated ten-year returns like a sample and calculate sample moments.

6.14 Review of Estimating Ten-Year Returns

- Let x_t be monthly returns.
- A ten-year return compounds 120 monthly returns:

$$R_{10\text{Y}} = \prod_{j=0}^{120-1} (1 + x_{t+j}) - 1.$$

- Assume monthly returns have a common mean $\text{E}(x_t)$, estimated by the sample average:

$$\bar{x} = \frac{1}{n} \sum_{t=1}^{n} x_t.$$

- We showed that if monthly returns are uncorrelated, the expected ten-year return can be estimated as

$$\bar{R}_{10\text{Y}} = (1 + \bar{x})^{120} - 1.$$

6.15 Monte Carlo Simulation

- The analytical approach gives a simple estimate:

$$\bar{R}_{10\text{Y}} = (1 + \bar{x})^{120} - 1.$$

- Monte Carlo simulation takes a different approach.
- Simulate many artificial ten-year return histories.
  - Each history contains 120 monthly returns.
  - Assume monthly returns are uncorrelated.
- For each simulation, calculate the ten-year return:

$$R_{10\text{Y}}^{(s)} = \prod_{j=0}^{12\times 10-1} (1 + x_j^{(s)}) - 1.$$

- Average across simulations to estimate the expected ten-year return.

6.16 Simulated 10-Year Return

```
VTI_mean = mean(ETF_data$VTI)
VTI_10Y_closed = (1 + VTI_mean)^(120) - 1

n_mc = 10000
VTI_MC = ETF_data %>%
  slice_sample(n = n_mc*120, replace = TRUE) %>%
  mutate(mc_idx = rep(1:n_mc, each = 120)) %>%
  group_by(mc_idx) %>%
  summarize(R10Y = prod(1 + VTI) - 1)
VTI_10Y_sim = mean(VTI_MC$R10Y)
```

```
##
## The close-form analytical 10-year VTI return is 214.376%.
##
## The simulated 10-year VTI return is 213.400%.
```

6.17 Ten-Year Return Variance

- Expected ten-year returns are manageable analytically.
- Variance is harder because returns are compounded:

$$R_{10Y} = \prod_{j=0}^{119}(1 + x_{t+j}) - 1.$$

- Expanding creates $2^{120} - 1$ product terms.
  - monthly returns: $x_t, x_{t+1}, \ldots$
  - two-month products: $x_t x_{t+1}, x_t x_{t+2}, \ldots$
  - all the way up to $x_t x_{t+1} \cdots x_{t+119}$
- Monte Carlo avoids managing this formula directly.

6.18 Simulated 10-Year Return VTI

```
VTI_10Y_mean = mean(VTI_MC$R10Y)
VTI_10Y_sd = sd(VTI_MC$R10Y)
VTI_10Y_rar = VTI_10Y_mean / VTI_10Y_sd
```

```
##
##
## VTI 10-year return
##
## Average return: 213.40%
##
## Standard deviation: 156.51%
##
## Risk-adjusted return: 1.3635
```

6.19 Simulated 10-Year Return 80/20 Portfolio

```
n_mc = 10000
Portfolio_MC = ETF_data %>%
  slice_sample(n = n_mc*120, replace = TRUE) %>%
  mutate(mc_idx = rep(1:n_mc, each = 120)) %>%
  group_by(mc_idx) %>%
  summarize(R10Y = prod(1 + 0.8 * VTI + 0.2 * AGG) - 1)
Portfolio_10Y_mean = mean(Portfolio_MC$R10Y)
Portfolio_10Y_sd = sd(Portfolio_MC$R10Y)
Portfolio_10Y_rar = Portfolio_10Y_mean / Portfolio_10Y_sd
```

```
##
##
## 80/20 Portfolio 10-year return
##
## Average return: 166.02%
##
## Standard deviation: 105.55%
##
## Risk-adjusted return: 1.5729
```

Lecture 7

Prediction and OLS

7.1 Prediction

- We have an outcome y we want to predict.
- y is a random variable that may depend on observed and unobserved factors.
- For observation i, a prediction function maps observed information into a prediction:

$$\hat{y}_i = g(x_i).$$

- The prediction error is

$$e_i = y_i - g(x_i).$$

- We evaluate predictions using squared error:

$$e_i^2 = (y_i - \hat{y}_i)^2 = (y_i - g(x_i))^2.$$

7.2 Estimating a Prediction Function

- We observe n pairs:

$$(y_1, x_1), (y_2, x_2), \ldots, (y_n, x_n).$$

- Choose a prediction function with unknown parameters:

$$g(x_i).$$

- Estimate the parameters by minimizing the sum of squared errors:

$$\text{SSE} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - g(x_i)\right)^2.$$

7.3 Ordinary Least Squares

- A common prediction function is linear:

$$g(x_i) = b_1 + b_2 x_i.$$

- Choose b_1 and b_2 to minimize the sum of squared errors:

$$\text{SSE} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - b_1 - b_2 x_i)^2.$$

- This estimation method is called *ordinary least squares*.
- OLS provides the best linear predictor.

7.4 OLS with Multiple Predictors

- In practice, we use many predictors to predict y.
- Let x_i be a $K \times 1$ vector:

$$x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iK} \end{bmatrix}.$$

- The prediction function becomes

$$g(x_i) = x_i' b,$$

where b is a $K \times 1$ vector of parameters.

7.5 Understanding the Prediction

- The prediction is a vector product:

$$\begin{aligned} \hat{y}_i = g(x_i) &= \begin{bmatrix} x_{i1} & x_{i2} & \cdots & x_{iK} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_K \end{bmatrix} \\ &= x_{i1}b_1 + x_{i2}b_2 + \cdots + x_{iK}b_K. \end{aligned}$$

- Typically, define $x_{i1} = 1$ as the constant term.
- Then b_1 is the intercept and the prediction when all other predictors equal zero:

$$\hat{y}_i = b_1 \qquad \text{when } x_{i2} = \cdots = x_{iK} = 0.$$

7.6 The OLS Estimator

- Choose b to minimize the sum of squared errors:

$$b = \underset{b}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - x_i' b\right)^2.$$

- Since

$$x_i' b = x_{i1}b_1 + x_{i2}b_2 + \cdots + x_{iK}b_K,$$

$$\text{SSE} = \sum_{i=1}^{n} \left(y_i - x_{i1}b_1 - x_{i2}b_2 - \cdots - x_{iK}b_K\right)^2.$$

- Take K partial derivatives and set them equal to zero:

$$\frac{\partial \text{SSE}}{\partial b_k} = -2 \sum_{i=1}^{n} x_{ik} \left(y_i - x_i' b\right) = 0, \qquad k = 1, \ldots, K.$$

7.7 The OLS Solution

- The OLS estimator chooses b to satisfy

$$\frac{\partial \text{SSE}}{\partial b_k} = -2\sum_{i=1}^{n} x_{ik}\left(y_i - x_i'b\right) = 0, \qquad k = 1, \ldots, K.$$

- Recall that the error is

$$e_i = y_i - x_i'b.$$

- Therefore, the first-order conditions imply

$$\sum_{i=1}^{n} x_{ik} e_i = 0, \qquad k = 1, \ldots, K.$$

7.8 Vectorizing the OLS Solution

- The K first-order conditions are

$$\sum_{i=1}^{n} x_{i1} e_i = 0, \quad \sum_{i=1}^{n} x_{i2} e_i = 0, \quad \ldots \quad \sum_{i=1}^{n} x_{iK} e_i = 0.$$

- Define $\mathbf{x}_k$ as the $n \times 1$ vector containing predictor k:

$$\mathbf{x}_k = \begin{bmatrix} x_{1k} \\ x_{2k} \\ \vdots \\ x_{nk} \end{bmatrix}, \qquad e = \begin{bmatrix} y_1 - x_1'b \\ y_2 - x_2'b \\ \vdots \\ y_n - x_n'b \end{bmatrix} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}.$$

- Each condition can be written as

$$\mathbf{x}_k' e = 0, \qquad k = 1, \ldots, K.$$

7.9 In Matrix Form

- Combine the K predictor vectors column-wise to create an $n \times K$ matrix:

$$X = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_K \end{bmatrix}.$$

- Y is the $n \times 1$ vector of outcomes:

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \qquad \hat{Y} = \begin{bmatrix} x_1'b \\ x_2'b \\ \vdots \\ x_n'b \end{bmatrix} = Xb.$$

- The error vector is

$$e = Y - \hat{Y} = Y - Xb.$$

- The OLS first-order conditions are

$$X'e = X'(Y - \hat{Y}) = X'(Y - Xb) = \mathbf{0}_{K \times 1}.$$

7.10 The OLS Normal Equations

- The first-order conditions are
$$X'(Y - Xb) = \mathbf{0}_{K\times 1}.$$
- Solve the system of equations:
$$\begin{aligned} \mathbf{0} &= X'(Y - Xb) \\ &= X'Y - X'Xb \\ X'Xb &= X'Y \\ (X'X)^{-1}(X'X)b &= (X'X)^{-1}X'Y \\ I_{K\times K}b &= (X'X)^{-1}X'Y \\ \boxed{b &= (X'X)^{-1}X'Y} \end{aligned}$$

7.11 OLS Example: MLB Data

- Analyze the 2023 season for 30 MLB teams.
- Let total season wins for team i be
$$\text{Wins}_i.$$
- Estimate the prediction function using OLS:
$$\widehat{\text{Wins}}_i = b_1 + b_2\text{R}_i + b_3\text{RA}_i.$$
- R_i is runs scored by team i.
- RA_i is runs allowed by team i.

7.12 OLS with MLB Data

- Build the outcome vector:
$$Y = \begin{bmatrix} \text{Wins}_1 \\ \text{Wins}_2 \\ \vdots \\ \text{Wins}_n \end{bmatrix}_{n\times 1}.$$
- Build the design matrix:
$$X = \begin{bmatrix} 1 & \text{R}_1 & \text{RA}_1 \\ 1 & \text{R}_2 & \text{RA}_2 \\ \vdots & \vdots & \vdots \\ 1 & \text{R}_n & \text{RA}_n \end{bmatrix}_{n\times 3}.$$
- Estimate the OLS coefficients:
$$b = (X'X)^{-1}X'Y.$$

7.13 Lahman Library

```
suppressMessages(library(dplyr))
suppressMessages(library(Lahman))

mlb_2023 <- Teams %>%
  filter(yearID == 2023) %>%
  select(teamID, name, W, R, RA)
```

```
##
## Data on 30 MLB teams in 2023
##
##    teamID                  name  W   R  RA
## 21    PHI Philadelphia Phillies 90 796 715
## 22    PIT    Pittsburgh Pirates 76 692 790
## 23    SDN      San Diego Padres 82 752 648
## 24    SEA      Seattle Mariners 88 758 659
## 25    SFN  San Francisco Giants 79 674 719
## 26    SLN   St. Louis Cardinals 71 719 829
## 27    TBA        Tampa Bay Rays 99 860 665
## 28    TEX         Texas Rangers 90 881 716
## 29    TOR     Toronto Blue Jays 89 746 671
## 30    WAS  Washington Nationals 71 700 845
```

7.14 OLS Estimates

```
n = nrow(mlb_2023)
Y = matrix(mlb_2023$W, ncol = 1)
X = as.matrix(cbind(rep(1, n),
                    mlb_2023$R,
                    mlb_2023$RA))

b = solve( t(X) %*% X) %*% (t(X) %*% Y)
```

```
##          b1          b2          b3
## 89.55410905  0.08284065 -0.09428070
```

7.15 Decomposing the Total Variation in y

- Given a prediction, decompose the outcome as

$$y_i = \underbrace{\hat{y}_i}_{\text{prediction}} + \underbrace{e_i}_{\text{prediction error}}.$$

- The Sum of Squares Total (SST) describes the total variation in y:

$$\text{SST} = \sum_{i=1}^{n} (y_i - \bar{y})^2.$$

- The Sum of Squared Errors (SSE) describes the variation in y unexplained by the prediction:

$$\text{SSE} = \sum_{i=1}^{n} e_i^2.$$

7.16 Goodness of Fit: R^2

- R^2 characterizes how well we fit or predict the outcome variable.
- R^2 is the fraction of sample variation in y explained by the prediction $\hat{y}$.
- Since SSE/SST is the fraction unexplained,

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} \in [0, 1].$$

- $R^2 = 0 \to 0\%$ of the variation in y is explained by $\hat{y}$.
- $R^2 = 1 \to 100\%$ of the variation in y is explained by $\hat{y}$.

7.17 R-Squared

```
err = Y - X%*%b
SSE = sum(err^2)
SST = sum((Y - mean(Y))^2)
r_squared = 1 - SSE/SST
```

```
##          SSE          SST    R-squared
##  538.5235663 4994.0000000    0.8921659
```

7.18 An Alternative MLB Predictor

- b_3 appears very similar to $-b_2$.
- Estimate the prediction function under the assumption $b_3 = -b_2$:

$$\begin{aligned}\widehat{\text{Wins}}_i &= b_1 + b_2\text{R}_i + b_3\text{RA}_i \\ &= b_1 + b_2\text{R}_i - b_2\text{RA}_i \\ &= b_1 + b_2(\text{R}_i - \text{RA}_i) \\ &= b_1 + b_2\text{RD}_i,\end{aligned}$$

  where RD_i is run differential.
- Do run levels matter, or only run differential?
  - Compare the R^2 between the two prediction functions.

7.19 Comparing the R-Squared

```
X_sm = as.matrix(cbind(rep(1, n),
                       mlb_2023$R - mlb_2023$RA))
b_sm = solve( t(X_sm) %*% X_sm) %*% (t(X_sm) %*% Y)
err_sm = Y - X_sm%*%b_sm

r_squared_sm = 1 - sum(err_sm^2)/SST
```

```
##          b1          b2
## 81.00000000  0.08852526
##
##  Original R-squared 0.89217
##
##  R-squared from smaller prediction equation 0.89071
```

Lecture 8

Linear Regression Model

8.1 Conditional Expectation

- $\mathrm{E}(y)$ is the unconditional expectation of y.
  - It summarizes the average value of y.
- The conditional expectation function summarizes how the expected value of y depends on x:

$$\mathrm{E}(y \mid x) \equiv m(x).$$

- Decompose each outcome as

$$y_i = m(x_i) + u_i.$$

- u_i is the component of y_i unexplained by x_i:

$$u_i = y_i - m(x_i).$$

- By construction, u_i has conditional mean zero:

$$\mathrm{E}(u_i \mid x_i) = \mathrm{E}(y_i - m(x_i) \mid x_i) = \mathrm{E}(y_i \mid x_i) - m(x_i) = 0.$$

8.2 The CEF as the Best Predictor

- What function $g(x)$ minimizes expected squared error?

$$\mathrm{E}\left[(y - g(x))^2\right].$$

- Since $y = m(x) + u$,

$$\begin{aligned}\mathrm{E}\left[(y - g(x))^2\right] &= \mathrm{E}\left[(m(x) + u - g(x))^2\right] \\ &= \mathrm{E}(u^2) + 2\,\mathrm{E}\left[u(m(x) - g(x))\right] + \mathrm{E}\left[(m(x) - g(x))^2\right].\end{aligned}$$

- The middle term is zero:

$$\mathrm{E}\left[u(m(x) - g(x))\right] = \mathrm{E}_x\left[(m(x) - g(x))\,\mathrm{E}(u \mid x)\right] = 0.$$

- Therefore,

$$\mathrm{E}\left[(y - g(x))^2\right] = \mathrm{E}(u^2) + \mathrm{E}\left[(m(x) - g(x))^2\right].$$

- This is minimized by setting $g(x) = m(x)$.

8.3 So What Exactly Does OLS Recover?

- The function that minimizes expected squared error is the conditional expectation function (CEF):
$$\mathrm{E}(y \mid x).$$
- OLS minimizes the sample sum of squared errors using a linear prediction function:
$$\hat{y}_i = x_i'b.$$
- If the CEF is linear,
$$\mathrm{E}(y_i \mid x_i) = x_i'\beta,$$
then b estimates β.
- Otherwise, $\hat{y}_i = x_i'b$ is best linear predictor given x_i.
- If x_i is flexible enough, $x_i'b$ can approximate $\mathrm{E}(y_i \mid x_i)$.

8.4 Recall the OLS Estimator

- Choose b to minimize the sum of squared errors:
$$\mathrm{SSE} = \sum_{i=1}^{n} \left(y_i - x_{i1}b_1 - x_{i2}b_2 - \cdots - x_{iK}b_K\right)^2.$$
- Take K partial derivatives and set them equal to zero:
$$\frac{\partial \mathrm{SSE}}{\partial b_k} = -2\sum_{i=1}^{n} x_{ik}\left(y_i - x_i'b\right) = 0, \qquad k = 1, \ldots, K.$$
- Given that the error is
$$e_i = y_i - x_i'b,$$
the first-order conditions imply
$$\sum_{i=1}^{n} x_{ik}e_i = 0, \qquad k = 1, \ldots, K.$$

8.5 Important Features of the OLS Solution

- If $x_{i1} = 1$ is a constant, then the FOCs imply
$$\sum_{i=1}^{n} x_{i1}e_i = 0 \quad \Rightarrow \quad \sum_{i=1}^{n} e_i = 0 \quad \Rightarrow \quad \bar{e} = 0.$$
- Since $e_i = y_i - x_i'b$,
$$\bar{e} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - x_i'b\right) = \bar{y} - \bar{x}'b.$$
- Therefore,
$$\bar{e} = 0 \quad \Rightarrow \quad \bar{y} = \bar{x}'b.$$
- The average error is zero and the prediction for the average x equals the average y.

8.6 Example: OLS in the Constant-Only Case

- If the prediction function includes only a constant:

$$\hat{y}_i = b_1.$$

- OLS has one first-order condition:

$$\sum_{i=1}^{n} e_i = 0 \qquad \Rightarrow \qquad \sum_{i=1}^{n} (y_i - b_1) = 0.$$

- Solving for b_1:

$$0 = \frac{1}{n} \sum_{i=1}^{n} (y_i - b_1) = \bar{y} - b_1.$$

$$\boxed{b_1 = \bar{y}}$$

- In this case, the OLS estimator is the sample average.

8.7 Matrix Form: Constant-Only Case

- In the constant-only case, the matrices are

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}, \qquad X = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{n \times 1}.$$

- Therefore,

$$X'X = n, \qquad X'Y = \sum_{i=1}^{n} y_i.$$

- Applying the OLS formula:

$$b = (X'X)^{-1} X'Y = n^{-1} \sum_{i=1}^{n} y_i = \bar{y}.$$

- Same answer from matrix algebra.

8.8 Example: Bivariate OLS

- For a constant and one predictor,

$$\hat{y}_i = b_1 + b_2 x_i.$$

- The first-order conditions imply

$$\sum_{i=1}^{n} e_i = 0, \qquad \sum_{i=1}^{n} x_i e_i = 0.$$

- Substituting $e_i = y_i - b_1 - b_2 x_i$ into the first condition and taking averages:

$$0 = \frac{1}{n}\sum_{i=1}^{n}(y_i - b_1 - b_2 x_i) = \bar{y} - b_1 - b_2\bar{x}.$$

$$\boxed{b_1 = \bar{y} - b_2\bar{x}}$$

8.9 Example: Bivariate OLS

- From the first-order condition for the constant,

$$b_1 = \bar{y} - b_2\bar{x}.$$

- Therefore,

$$e_i = y_i - b_1 - b_2 x_i = y_i - \bar{y} - b_2(x_i - \bar{x}).$$

- Plugging into the first-order condition:

$$\sum_{i=1}^{n} x_i \left[y_i - \bar{y} - b_2(x_i - \bar{x})\right] = 0.$$

- Solving for b_2,

$$\boxed{b_2 = \frac{\sum_{i=1}^{n} x_i(y_i - \bar{y})}{\sum_{i=1}^{n} x_i(x_i - \bar{x})} = \frac{\widehat{\mathrm{Cov}}(x,y)}{\widehat{\mathrm{Var}}(x)}}$$

8.10 Matrix Form: Bivariate OLS

- In the bivariate OLS case, the matrices are

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1}, \qquad X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}_{n\times 2}.$$

- Applying the OLS formula:

$$b = (X'X)^{-1}X'Y = \begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix}.$$

- Using sample averages and factoring out n

$$b = \begin{bmatrix} n & n\bar{x} \\ n\bar{x} & n\overline{x^2} \end{bmatrix}^{-1} \begin{bmatrix} n\bar{y} \\ n\overline{xy} \end{bmatrix} = \begin{bmatrix} 1 & \bar{x} \\ \bar{x} & \overline{x^2} \end{bmatrix}^{-1} \begin{bmatrix} \bar{y} \\ \overline{xy} \end{bmatrix}.$$

8.11 Matrix Form: Bivariate OLS

- From the previous slide,
$$b = \begin{bmatrix} 1 & \bar{x} \\ \bar{x} & \overline{x^2} \end{bmatrix}^{-1} \begin{bmatrix} \bar{y} \\ \overline{xy} \end{bmatrix}.$$
- The inverse is
$$\begin{bmatrix} 1 & \bar{x} \\ \bar{x} & \overline{x^2} \end{bmatrix}^{-1} = \frac{1}{\overline{x^2} - \bar{x}^2} \begin{bmatrix} \overline{x^2} & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix}.$$
- Focus on the second element of b:
$$b_2 = \frac{1}{\overline{x^2} - \bar{x}^2}(-\bar{x}\bar{y} + \overline{xy}) = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2}$$
$$\boxed{b_2 = \frac{\widehat{\mathrm{Cov}}(x, y)}{\widehat{\mathrm{Var}}(x)}}$$

8.12 Example: Bivariate Normal

- Suppose y_i and x_i are bivariate normal:
$$\begin{bmatrix} y_i \\ x_i \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_y \\ \mu_x \end{bmatrix}, \begin{bmatrix} \sigma_y^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_x^2 \end{bmatrix} \right).$$
- Multivariate normal random variables have linear conditional expectation functions:
$$\mathrm{E}(y_i \mid x_i) = \mu_y + \frac{\sigma_{xy}}{\sigma_x^2}(x_i - \mu_x).$$
- Rearranging,
$$\mathrm{E}(y_i \mid x_i) = \left(\mu_y - \frac{\sigma_{xy}}{\sigma_x^2}\mu_x \right) + \left(\frac{\sigma_{xy}}{\sigma_x^2} \right) x_i.$$
- Since the CEF is linear, OLS estimates its slope and intercept.

8.13 Econometric Modeling in Data Analysis

- A model is a set of assumptions about the joint distribution of the variables we analyze.
  - Equivalently, a model places restrictions on the data-generating process.
- These assumptions can involve both observed and unobserved variables.
- The objective is to characterize relationships between variables.
- The model tells us how to analyze the data and how to interpret the results.

8.14 Linear Regression Model

- The *Linear Regression Model (LRM)* is the most widely applied econometric tool.
  - Defined by five assumptions on observed and unobserved variables.
- The LRM relates an outcome y to explanatory variables $x_1, x_2, \ldots, x_K$:

$$y = f(x_1, x_2, \ldots, x_K) + \varepsilon.$$

- ε is the *disturbance*.
  - It contains all unobserved determinants of y.
- We observe y and the x's, but not ε.
- The goal is to learn about $f(\cdot)$.

8.15 Assumption 1: Linearity

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \varepsilon_i.$$

- Usually set $x_{i1} = 1$ to include a constant term.
- Linearity means *linear in parameters*.

$$y_i = \beta_1 f_1(x_i) + \beta_2 f_2(x_i) + \cdots + \beta_K f_K(x_i) + \varepsilon_i.$$

- $f_k(x_i)$ can be any known function of the explanatory variables.
- Valid: $\ln(y_i) = \beta_1 + \beta_2 \sin(x_{i1} + x_{i2}) + \varepsilon_i$.
- Invalid: $y_i = \beta_1 + x_i^{\beta_2} + \varepsilon_i$.

8.16 Data Organization Notation

- Each individual observation satisfies

$$y_i = x_i'\beta + \varepsilon_i, \qquad x_i = \begin{bmatrix} 1 \\ x_{i2} \\ \vdots \\ x_{iK} \end{bmatrix}_{K\times 1}, \qquad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{bmatrix}_{K\times 1}$$

- For n observations, stack the data:

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1}, \qquad X = \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix}_{n\times K}, \qquad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n\times 1}$$

- In matrix form,

$$Y = X\beta + \boldsymbol{\varepsilon}.$$

8.17 Assumption 2: Full Rank of X

- Let $\mathbf{x}_k$ be the $n \times 1$ vector containing variable k:

$$\mathbf{x}_k = \begin{bmatrix} x_{1k} \\ x_{2k} \\ \vdots \\ x_{nk} \end{bmatrix}, \qquad k = 1, 2, \ldots, K.$$

- The design matrix is

$$X = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_K \end{bmatrix}.$$

- Full rank means no $\mathbf{x}_k$ is a perfect linear combination of the other columns.
- Nonlinear transformations are allowed:

$$\mathbf{x}_k = \mathbf{x}_j^2.$$

- This assumption ensures $X'X$ is invertible.

8.18 Assumption 3: Mean Independence

- The disturbance is mean independent of the explanatory variables:

$$\mathrm{E}(\varepsilon_i \mid X) = \mathrm{E}(\varepsilon_i) = 0, \qquad i = 1, \ldots, n.$$

- Viewing this assumption in terms of prediction, the best predictor of ε_i does not depend on X.
- Equivalently, knowing X does not change the average value of the disturbance.
- This is not an assumption about $\mathrm{Var}(\varepsilon_i \mid X)$ or any other part of the distribution of ε_i

8.19 Estimating the Linear Regression Model

- Assumption 1, linearity, implies

$$\mathrm{E}(y_i \mid x_i) = \mathrm{E}(x_i'\beta + \varepsilon_i \mid x_i) = x_i'\beta + \mathrm{E}(\varepsilon_i \mid x_i).$$

- Combined with Assumption 3, mean independence,

$$\mathrm{E}(\varepsilon_i \mid x_i) = 0,$$

the conditional expectation function is linear:

$$\mathrm{E}(y_i \mid x_i) = x_i'\beta.$$

- Therefore, β can be estimated with OLS:

$$b = (X'X)^{-1}X'Y.$$

- Assumption 2, full rank, ensures $X'X$ is invertible.

8.20 Summary

- Under Assumptions 1–3 of the Linear Regression Model, y_i has a linear conditional expectation function:
$$\mathrm{E}(y_i \mid x_i) = x_i'\beta.$$
- The model can be estimated with OLS:
$$b = (X'X)^{-1}X'Y.$$
- OLS estimates the linear CEF because it minimizes the SSE using a linear prediction function.
- We use the word *regression* to mean conditional expectation, which is where the model gets its name.
- Assumptions 4–5 are needed for inference.
  - Since b is a random variable, we need its variance and distribution.

Lecture 9

Properties of OLS

9.1 We Proposed OLS to Estimate the LRM

- The Linear Regression Model is defined by assumptions on the data and parameters β:

$$Y = X\beta + \boldsymbol{\varepsilon}.$$

- We proposed estimating β with OLS:

$$b = (X'X)^{-1}X'Y.$$

- Assumptions 1–3 of the LRM imply y_i has a linear conditional expectation function:

$$\mathrm{E}(y_i \mid x_i) = x_i'\beta.$$

- Since OLS estimates linear prediction functions, it is a natural estimator of the linear CEF.

9.2 Is b the Best Estimator for β?

- b is a random variable because it is a function of random data.
- To evaluate OLS, we need to study its sampling properties.
- We would like to show that b is:

1) *Unbiased*: $\mathrm{E}(b) = \beta$.
2) *Consistent*: $b \to \beta$ as $n \to \infty$.
3) *Efficient*: $\mathrm{Var}(b)$ is small relative to other estimators.
4) *Approximately Normally Distributed*.

9.3 Rewriting the OLS Estimator

- $b = (X'X)^{-1}X'Y$ estimates β in the model

$$Y = X\beta + \boldsymbol{\varepsilon}.$$

- Substituting the model into the estimator:

$$\begin{aligned} b &= (X'X)^{-1}X'Y \\ &= (X'X)^{-1}X'(X\beta + \boldsymbol{\varepsilon}) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\boldsymbol{\varepsilon} \\ &= \beta + A\boldsymbol{\varepsilon}, \end{aligned}$$

  where

$$A = (X'X)^{-1}X'.$$

- Therefore,

$$b = \beta + A\boldsymbol{\varepsilon}.$$

9.4 The Determinants of Randomness in b

- The OLS estimator is
$$b = \beta + A\boldsymbol{\varepsilon}, \qquad A = (X'X)^{-1}X'.$$
- The estimator equals β plus a vector of random variables, $A\boldsymbol{\varepsilon}$.
  - $A\boldsymbol{\varepsilon}$ is a linear function of the disturbances ε_i.
- The sampling distribution of b depends on the distribution of $A\boldsymbol{\varepsilon}$.
- Interestingly, $A\boldsymbol{\varepsilon}$ can itself be interpreted as a vector of OLS coefficients:
$$(X'X)^{-1}X'\boldsymbol{\varepsilon}$$
$\Rightarrow$ OLS coefficients from regressing $\boldsymbol{\varepsilon}$ on X.

9.5 Properties of OLS: b is Unbiased

- Given
$$b = \beta + A\boldsymbol{\varepsilon}, \qquad A = (X'X)^{-1}X'.$$
- To show that b is unbiased, we need
$$\mathrm{E}(b) = \mathrm{E}(\beta + A\boldsymbol{\varepsilon}) = \beta + \mathrm{E}(A\boldsymbol{\varepsilon}) = \beta.$$
- Therefore, it is sufficient to show
$$\mathrm{E}(A\boldsymbol{\varepsilon}) = 0.$$

9.6 Showing $\mathrm{E}(A\boldsymbol{\varepsilon}) = 0$

- First, condition on X. Since A is a function of X, it is fixed conditional on X:
$$\begin{aligned}\mathrm{E}(A\boldsymbol{\varepsilon} \mid X) &= \mathrm{E}\left[(X'X)^{-1}X'\boldsymbol{\varepsilon} \mid X\right] \\ &= (X'X)^{-1}X'\,\mathrm{E}(\boldsymbol{\varepsilon} \mid X) \\ &= 0 \quad \boxed{\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0 \text{ by ASM 3.}}\end{aligned}$$
- Then apply the law of iterated expectations:
$$\mathrm{E}(A\boldsymbol{\varepsilon}) = \mathrm{E}_X\left[\mathrm{E}(A\boldsymbol{\varepsilon} \mid X)\right] = 0.$$

9.7 Proving OLS is Unbiased

- Under Mean Independence, b is an unbiased estimator of β:
$$\mathrm{E}(b) = \beta.$$
- Given
$$b = (X'X)^{-1}X'Y = (X'X)^{-1}X'(X\beta + \boldsymbol{\varepsilon}).$$

- Taking the conditional expectation,

$$\begin{aligned} \mathrm{E}(b \mid X) &= \mathrm{E}\left[(X'X)^{-1}X'(X\beta + \boldsymbol{\varepsilon}) \mid X\right] \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\,\mathrm{E}(\boldsymbol{\varepsilon} \mid X) \\ &= \beta \qquad\qquad \boxed{\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0 \text{ by ASM 3.}} \end{aligned}$$

- Applying the law of iterated expectations,

$$\mathrm{E}(b) = \mathrm{E}_X\left[\mathrm{E}(b \mid X)\right] = \mathrm{E}_X(\beta) = \beta.$$

9.8 Properties of OLS: Variance of b

- Since b is a $K \times 1$ random vector, $\mathrm{Var}(b)$ is a $K \times K$ variance-covariance matrix:

$$\mathrm{Var}(b) = \begin{bmatrix} \mathrm{Var}(b_1) & \mathrm{Cov}(b_1, b_2) & \cdots & \mathrm{Cov}(b_1, b_K) \\ \mathrm{Cov}(b_2, b_1) & \mathrm{Var}(b_2) & \cdots & \mathrm{Cov}(b_2, b_K) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(b_K, b_1) & \mathrm{Cov}(b_K, b_2) & \cdots & \mathrm{Var}(b_K) \end{bmatrix}.$$

- The diagonal elements are the variances of the individual OLS coeffients.
- The off-diagonal elements are the covariances between OLS coeffients.
  - The covariances are non-zero since they are estimated with the same data.

9.9 Variance of b

- Recall

$$b = \beta + A\boldsymbol{\varepsilon}, \qquad A = (X'X)^{-1}X'.$$

- Therefore,

$$\mathrm{Var}(b) = \mathrm{Var}(\beta + A\boldsymbol{\varepsilon}) = \mathrm{Var}(A\boldsymbol{\varepsilon}).$$

- Thus, the variability in b comes from the variability in $A\boldsymbol{\varepsilon}$.
- Since A is a function of X, conditioning on X treats A as fixed:

$$\mathrm{Var}(b \mid X) = \mathrm{Var}(A\boldsymbol{\varepsilon} \mid X) = A\,\mathrm{Var}(\boldsymbol{\varepsilon} \mid X)A'.$$

- Therefore, $\mathrm{Var}(b \mid X)$ depends on the variance-covariance matrix of the disturbances.

9.10 Variance-Covariance of the Disturbances

- Since $\boldsymbol{\varepsilon}$ is an $n \times 1$ vector of disturbances,

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 & \varepsilon_2 & \ldots & \varepsilon_n \end{bmatrix}',$$

  $\mathrm{Var}(\boldsymbol{\varepsilon})$ is an $n \times n$ variance-covariance matrix:

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \begin{bmatrix} \mathrm{Var}(\varepsilon_1) & \mathrm{Cov}(\varepsilon_1, \varepsilon_2) & \cdots & \mathrm{Cov}(\varepsilon_1, \varepsilon_n) \\ \mathrm{Cov}(\varepsilon_2, \varepsilon_1) & \mathrm{Var}(\varepsilon_2) & \cdots & \mathrm{Cov}(\varepsilon_2, \varepsilon_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(\varepsilon_n, \varepsilon_1) & \mathrm{Cov}(\varepsilon_n, \varepsilon_2) & \cdots & \mathrm{Var}(\varepsilon_n) \end{bmatrix}.$$

- The diagonal elements are the variances of the individual disturbances.
- The off-diagonal elements are the covariances between disturbances.

9.11 Variance-Covariance of the Disturbances

- To derive $\text{Var}(b \mid X)$, we need assumptions about the structure of $\text{Var}(\boldsymbol{\varepsilon} \mid X)$.
- LRM Assumption 4: Homoskedasticity and non-autocorrelation.

1) Homoskedasticity:
$$\text{Var}(\varepsilon_i \mid X) = \sigma^2.$$
   - The variance is constant across disturbances and does not depend on X.

2) Non-autocorrelation:
$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \qquad i \neq j.$$
   - No linear co-movement between different disturbances.

9.12 $\text{Var}(\boldsymbol{\varepsilon} \mid X)$ Under Classical LRM Assumptions

- The conditional variance-covariance matrix is
$$\text{Var}(\boldsymbol{\varepsilon} \mid X) = \begin{bmatrix} \text{Var}(\varepsilon_1 \mid X) & \text{Cov}(\varepsilon_1, \varepsilon_2 \mid X) & \cdots & \text{Cov}(\varepsilon_1, \varepsilon_n \mid X) \\ \text{Cov}(\varepsilon_2, \varepsilon_1 \mid X) & \text{Var}(\varepsilon_2 \mid X) & \cdots & \text{Cov}(\varepsilon_2, \varepsilon_n \mid X) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\varepsilon_n, \varepsilon_1 \mid X) & \text{Cov}(\varepsilon_n, \varepsilon_2 \mid X) & \cdots & \text{Var}(\varepsilon_n \mid X) \end{bmatrix}.$$
- Under homoskedasticity,
$$\text{Var}(\varepsilon_i \mid X) = \sigma^2.$$
- Under non-autocorrelation, for $i \neq j$,
$$\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0.$$
- Therefore,
$$\boxed{\text{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I_{n \times n}}.$$

9.13 Variance of b

- The conditional variance is straightforward:
$$\text{Var}(b \mid X) = \text{Var}(\beta + A\boldsymbol{\varepsilon} \mid X) = A \,\text{Var}(\boldsymbol{\varepsilon} \mid X) A'.$$
- Using Assumption 4, $\text{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I_{n \times n}$:
$$\begin{aligned} \text{Var}(b \mid X) &= A\sigma^2 I_{n \times n} A' \\ &= \sigma^2 (X'X)^{-1} X'X (X'X)^{-1} \\ &= \sigma^2 (X'X)^{-1}. \end{aligned}$$

- By the law of total variance,
$$\mathrm{Var}(b) = \sigma^2 \,\mathrm{E}\left[(X'X)^{-1}\right].$$
  - The unconditional variance averages over the randomness in X.
- Estimating $\mathrm{E}[(X'X)^{-1}]$ with its sample realization $(X'X)^{-1}$ brings us back to $\mathrm{Var}(b \mid X)$

9.14 Properties of OLS: b is Efficient

- The Gauss-Markov Theorem provides an important justification for using OLS to estimate the LRM.
- Theorem: Under the assumptions of the linear regression model, OLS is the minimum-variance linear unbiased estimator of β.
  - Among all linear unbiased estimators, OLS has the smallest variance.
  - Linear means that the estimator is a linear function of Y.
- OLS is therefore *efficient* within the class of linear unbiased estimators.

9.15 Implications of the Gauss-Markov Theorem

- If the disturbances are heteroskedastic, OLS can remain unbiased but is no longer guaranteed to be efficient.
- Biased estimators can have smaller variance than OLS.
  - Gauss-Markov only compares linear unbiased estimators.
- The Gauss-Markov Theorem can also be used to establish the efficiency of other estimators.
  - Rewrite or transform the problem so that the estimator is OLS in a model satisfying the LRM assumptions.

9.16 OLS is a Function of a Sample Average

- Recall that x_i is a $K \times 1$ vector of explanatory variables and ε_i is a disturbance.
- Define $w_i = x_i\varepsilon_i$, a $K \times 1$ random vector with sample average
$$\bar{w} = \frac{1}{n}\sum_{i=1}^{n} w_i = \frac{1}{n}\sum_{i=1}^{n} x_i\varepsilon_i = \frac{X'\boldsymbol{\varepsilon}}{n}.$$
- OLS can be written as a function of this sample average:
$$\begin{aligned} b &= \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon} \\ &= \beta + \left(\frac{X'X}{n}\right)^{-1}\left(\frac{X'\boldsymbol{\varepsilon}}{n}\right) \\ &= \beta + \left(\frac{X'X}{n}\right)^{-1}\bar{w}. \end{aligned}$$

9.17 Probability Limit

- Let $b^{(n)}$ denote an estimator of β based on the first n observations.
- As the sample size grows,
$$b^{(1)}, b^{(2)}, \ldots, b^{(m)}, b^{(m+1)}, \ldots$$
forms a sequence of random variables based on increasingly large samples.
- We would like this sequence to converge to the true parameter:
$$\operatorname*{plim}_{n\to\infty} b^{(n)} = \beta.$$
- Define $D(b^{(n)}, \beta)$ as the distance between $b^{(n)}$ and β. For any $\eta > 0$,
$$\lim_{n\to\infty} \Pr\Big(D(b^{(n)}, \beta) \geq \eta\Big) = 0.$$

9.18 Large-Sample Properties of Averages

- Sample averages have two important large-sample properties.
- Law of Large Numbers:
$$\operatorname{plim} \bar{x} = \mathrm{E}(x).$$
  - As n grows, the sample average converges to the population mean.
- Central Limit Theorem:
  - As n grows, the sampling distribution of the sample average becomes approximately normal.
- Since the OLS estimator is a function of sample averages, it has analogous large-sample properties.

9.19 Properties of OLS: b is Consistent

- Recall
$$b = \beta + \left(\frac{X'X}{n}\right)^{-1} \bar{w}, \qquad \bar{w} = \frac{1}{n}\sum_{i=1}^{n} x_i\varepsilon_i.$$
- By Mean Independence,
$$\mathrm{E}(w_i) = \mathrm{E}(x_i\varepsilon_i) = 0.$$
- By the Law of Large Numbers,
$$\operatorname{plim} \bar{w} = 0, \qquad \operatorname{plim} \frac{X'X}{n} = \mathrm{E}(x_i x_i'),$$
- Therefore,
$$\operatorname{plim} b = \beta + \mathrm{E}(x_i x_i')^{-1} \operatorname{plim} \bar{w} = \beta.$$
- OLS is *consistent*.

9.20 Properties of OLS: b is Normally Distributed

Assumption 5 (CLRM): Normality of the disturbances:

$$\boldsymbol{\varepsilon} \mid X \sim N\left(0, \sigma^2 I_{n\times n}\right).$$

- Recall
$$b = \beta + A\boldsymbol{\varepsilon}, \qquad A = (X'X)^{-1}X'.$$
- A linear function of a normally distributed random vector is also normally distributed.

$$\boxed{b \mid X \sim N\left(\beta, \sigma^2 (X'X)^{-1}\right)}$$

9.21 Relaxing Assumption 5

- In practice, the distribution of the disturbances is generally unknown.
- Assuming normality is needed for exact finite-sample normality of b, but not for large-sample inference.
- Recall that OLS is a function of a sample average:
$$b = \beta + \left(\frac{X'X}{n}\right)^{-1} \bar{w}.$$
- By the Central Limit Theorem, $\bar{w}$ is approximately normally distributed in large samples.
- Therefore, b is approximately normally distributed in large samples.

Lecture 10

Interpretation of OLS

10.1 Interpreting LRM Parameters

- In the linear regression model

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \varepsilon_i.$$

- Each parameter β_k is intended to represent a causal effect.
- A *causal effect* describes how one variable affects another, holding everything else fixed.
- In the LRM, this effect is

$$\frac{\partial y_i}{\partial x_{ik}} = \beta_k.$$

  - In economics, this is often called a *marginal effect*.

10.2 Interpreting OLS Coefficients

- Can we interpret OLS coefficients as causal effects?
  - If Assumptions 1–3 hold, then yes.
- Under Assumptions 1–3, the conditional expectation function identifies the causal effects:

$$\mathrm{E}(y_i \mid x_i) = x_i'\beta.$$

  - The effect of x_{ik} on expected y_i is β_k.
- If Mean Independence fails:
  - The conditional expectation function may no longer be linear.
  - The parameters of the CEF diverge from β
- In this case, how do the OLS estimates relate to β?

10.3 OLS Coefficients Are Causal Effect-Like

- In the linear regression model: $y_i = x_i'\beta + \varepsilon_i$
- A causal effect has a very specific interpretation:
  - β_k is the effect of x_{ik} on y_i, holding everything else fixed.
- The OLS estimate b_k holds some things fixed, but not everything.
  - It holds the other variables in X fixed.
  - It does not hold the factors in ε_i fixed.
- Unless these factors are added to X, OLS cannot directly hold them fixed.
  - Instead, we rely on the Mean Independence assumption.

10.4 The Mechanics of OLS

- Separate the columns of X into two groups, X_1 and X_2.
- The Linear Regression Model is

$$Y = X_1\beta_1 + X_2\beta_2 + \boldsymbol{\varepsilon}.$$

  - β_1 represents the causal effects of X_1 on Y, holding X_2 and $\boldsymbol{\varepsilon}$ fixed.

- The OLS fitted values are

$$\hat{Y} = X_1 b_1 + X_2 b_2.$$

  - b_1 measures the effects of X_1 on predicted Y, holding X_2 fixed.

- How does OLS estimate an effect while "holding the other variables fixed"?

10.5 The Projection Matrix for X

- The projection matrix for X is

$$\mathbf{P}_X = X(X'X)^{-1}X'.$$

- Multiplying a vector by $\mathbf{P}_X$ projects it onto the column space of X.
- Thus, $\mathbf{P}_X Y$ is the projection of Y onto the column space of X.
  - It gives the values of Y predicted by a linear combination of the columns of X.

$$\mathbf{P}_X Y = X(X'X)^{-1}X'Y = Xb = \hat{Y}.$$

- Also,

$$\mathbf{P}_X X = X,$$

  because the columns of X can be predicted perfectly using X.

10.6 The Residual Maker for X

- The residual maker (or annihilator matrix) for X is

$$\mathbf{M}_X = I_{n\times n} - \mathbf{P}_X.$$

- Multiplying a vector by $\mathbf{M}_X$ removes the component that can be predicted by a linear combination of the columns of X.
- Thus, $\mathbf{M}_X Y$ gives the residuals from regressing Y on X:

$$\mathbf{M}_X Y = (I_{n\times n} - \mathbf{P}_X)Y = Y - \hat{Y} = e.$$

- Also,

$$\mathbf{M}_X X = 0,$$

  because nothing remains after removing from X the component predicted by X.

10.7 Partialling Out the Other Regressors

- OLS works by partialling out. Begin with:

$$Y = X_1\beta_1 + X_2\beta_2 + \boldsymbol{\varepsilon}.$$

- Define the residual maker for X_2:

$$\mathbf{M}_2 = I_{n\times n} - X_2(X_2'X_2)^{-1}X_2'.$$

- Multiply the model by $\mathbf{M}_2$:

$$\begin{aligned}\mathbf{M}_2Y &= \mathbf{M}_2X_1\beta_1 + \mathbf{M}_2X_2\beta_2 + \mathbf{M}_2\boldsymbol{\varepsilon}\\ &= \mathbf{M}_2X_1\beta_1 + \mathbf{M}_2\boldsymbol{\varepsilon}.\end{aligned}$$

- The variation explained by X_2 has been removed from Y and X_1.

10.8 Residualized Y and X_1

- Y can be decomposed into the component predicted by X_2 and the component that cannot be predicted:

$$Y = \mathbf{P}_2Y + \mathbf{M}_2Y = \hat{Y} + Y^*.$$

- Likewise, for X_1:

$$X_1 = \mathbf{P}_2X_1 + \mathbf{M}_2X_1 = \hat{X}_1 + X_1^*.$$

- The partialled-out model has the same parameter β_1, but uses the residualized variables:

$$Y^* = X_1^*\beta_1 + \mathbf{M}_2\boldsymbol{\varepsilon}.$$

10.9 How OLS Holds Other Regressors Fixed

- Start with $Y = X_1\beta_1 + X_2\beta_2 + \boldsymbol{\varepsilon}$.
- Partial out X_2: $Y^* = X_1^*\beta_1 + \boldsymbol{\varepsilon}^*$, where

$$Y^* = \mathbf{M}_2Y, \qquad X_1^* = \mathbf{M}_2X_1, \qquad \boldsymbol{\varepsilon}^* = \mathbf{M}_2\boldsymbol{\varepsilon}.$$

- Estimate b_1 by minimizing SSE in the residual-on-residual regression:

$$b_1 = \left[(X_1^*)'X_1^*\right]^{-1}(X_1^*)'Y^*.$$

- The effect of X_1, holding X_2 fixed, is estimated by:

  1) Removing the variation in Y and X_1 explained by X_2.
  2) Relating the remaining variation in X_1 to the remaining variation in Y.

10.10 OLS Not Holding X_2 Fixed

- Consider $Y = X_1\beta_1 + X_2\beta_2 + \boldsymbol{\varepsilon}$.
- Estimating OLS without holding X_2 fixed gives

$$b_1 = (X_1'X_1)^{-1}X_1'Y.$$

- Substituting the model for Y,

$$b_1 = (X_1'X_1)^{-1}X_1'(X_1\beta_1 + X_2\beta_2 + \boldsymbol{\varepsilon}).$$

- Distributing and simplifying,

$$b_1 = \beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2 + (X_1'X_1)^{-1}X_1'\boldsymbol{\varepsilon}.$$

10.11 Decomposing the OLS Estimate

- Consider the OLS coefficients from predicting X_2 and $\boldsymbol{\varepsilon}$ with X_1:

$$\Gamma = (X_1'X_1)^{-1}X_1'X_2, \qquad \hat{X}_2 = X_1\Gamma$$
$$\delta = (X_1'X_1)^{-1}X_1'\boldsymbol{\varepsilon}, \qquad \hat{\boldsymbol{\varepsilon}} = X_1\delta$$

- The OLS estimator can now be written as

$$b_1 = \beta_1 + \Gamma\beta_2 + \delta.$$

- The difference between b_1 and β_1 depends on:
  - The relationship between X_1 and X_2: Γ.
  - The effect of X_2 on Y: β_2.
  - The relationship between X_1 and $\boldsymbol{\varepsilon}$: δ.

10.12 Bias From Not Holding X_2 Fixed

- From the previous slide,

$$b_1 = \beta_1 + \Gamma\beta_2 + \delta.$$

- Taking expectations,

$$\mathrm{E}(b_1) = \beta_1 + \mathrm{E}(\Gamma)\beta_2 + \mathrm{E}(\delta).$$

- Assuming Mean Independence, $\mathrm{E}(\delta) = 0$.
- Thus, the bias from omitting X_2 is

$$\mathrm{E}(b_1 - \beta_1) = \mathrm{E}(\Gamma)\beta_2.$$

- Omitting X_2 introduces no bias if either:
  - X_2 has no effect on Y: $\beta_2 = 0$.
  - X_1 cannot predict X_2: $\mathrm{E}(\Gamma) = 0$.

10.13 Bias in OLS Estimates

- Omitting relevant variables from a regression can be a major source of bias.
- In the estimated model
$$Y = X\beta + \boldsymbol{\varepsilon},$$
the OLS estimator
$$b = (X'X)^{-1}X'Y$$
holds the variables in X fixed, but not the factors in $\boldsymbol{\varepsilon}$.
- If $\boldsymbol{\varepsilon}$ contains omitted variables X_2 with marginal effects β_2 on Y, then omitting them can introduce bias:
$$\mathrm{E}(b - \beta) = \mathrm{E}(\Gamma)\beta_2.$$
  - Γ measures how well X predicts X_2.

10.14 Signing the Bias

- If X_2 is believed to cause bias in b, we should include it in the regression.
- In general, we aim to include as many relevant variables as possible to ensure they are held fixed by OLS.
- If X_2 is unobserved, we can use
$$\mathrm{E}(b - \beta) = \mathrm{E}(\Gamma)\beta_2$$
to attempt to determine the direction of bias in the OLS estimates.

10.15 Bias with a Single Omitted Variable

- Assume
$$Y = X\beta + Z\alpha + \boldsymbol{\varepsilon}$$
satisfies the LRM assumptions, where Z is a single omitted variable.
- Predicting Z using X gives
$$\hat{z}_i = x_i'\gamma = x_{i1}\gamma_1 + \cdots + x_{iK}\gamma_K.$$
- The bias in each OLS coefficient is
$$\mathrm{E}(b_k - \beta_k) = \mathrm{E}(\gamma_k)\alpha.$$
- The sign of the bias depends on the signs of α and $\mathrm{E}(\gamma_k)$.
  - α: Effect of Z on Y.
  - γ_k: Effect of x_k on predicted Z, holding the other regressors fixed.

10.16 When Does Mean Independence Fail?

- If Mean Independence fails, the expected OLS coefficients are

$$\mathrm{E}(b) = \beta + \delta, \qquad \delta = \mathrm{E}\left[(X'X)^{-1}X'\boldsymbol{\varepsilon}\right].$$

- Mean Independence requires that the explanatory variables do not predict the average disturbance:

$$\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0.$$

  - If this fails, X contains information about the average value of $\boldsymbol{\varepsilon}$.

- Common reasons why Mean Independence fails:

  - Omitted relevant variables.
  - Simultaneity: X and Y are jointly determined.
  - Measurement error in X.

10.17 Classical Measurement Error

- Assume the correct model is

$$y_i = \beta_1 + \beta_2 x_i^{\mathrm{o}} + \varepsilon_i.$$

- But only a noisy measure of x_i^{o} is observed:

$$x_i = x_i^{\mathrm{o}} + u_i.$$

- Classical measurement error assumes u_i is independent of x_i^{o}.
- Although independent of x_i^{o}, measurement error is positively correlated with the observed x_i:

$$\mathrm{Cov}(x_i, u_i) = \mathrm{Var}(u_i) > 0.$$

10.18 Bias with Classical Measurement Error

- Given

$$y_i = \beta_1 + \beta_2 x_i^{\mathrm{o}} + \varepsilon_i.$$

- Substituting $x_i^{\mathrm{o}} = x_i - u_i$ into the model,

$$\begin{aligned} y_i &= \beta_1 + \beta_2(x_i - u_i) + \varepsilon_i \\ &= \beta_1 + \beta_2 x_i - \beta_2 u_i + \varepsilon_i. \end{aligned}$$

- If we estimate the regression using the observed x_i,

$$\hat{y}_i = b_1 + b_2 x_i,$$

  the measurement error u_i acts as an omitted variable with marginal effect $-\beta_2$.

- Thus, OLS is biased:

$$\mathrm{E}(b_2) \neq \beta_2.$$

10.19 Attenuation Bias

- Given
$$y_i = \beta_1 + \beta_2 x_i - \beta_2 u_i + \varepsilon_i.$$
- Consider predicting measurement error using x_i:
$$\hat{u}_i = \gamma_1 + \gamma_2 x_i, \qquad \gamma_2 = \frac{\widehat{\text{Cov}(x_i, u_i)}}{\widehat{\text{Var}(x_i)}}.$$
- Applying the OVB formula when mismeasured x_i is used,
$$\text{E}(b_2) = \beta_2 - \text{E}(\gamma_2)\beta_2 = [1 - \text{E}(\gamma_2)]\beta_2.$$
- Under classical measurement error, $0 < \text{E}(\gamma_2) < 1$.
  - Thus, OLS estimates are biased toward zero: *attenuation bias*.

Lecture 11

Inference with OLS

11.1 Sampling Distribution of OLS

- Under the assumptions of the linear regression model, the OLS estimator is

$$b = (X'X)^{-1}X'Y.$$

- With n observations and K parameters, the sampling distribution of coefficient b_k is

$$b_k \mid X \sim N\left(\beta_k, \text{Var}(b_k \mid X)\right).$$

- Standardizing b_k gives

$$\frac{b_k - \beta_k}{\sqrt{\text{Var}(b_k \mid X)}} \sim N(0,1).$$

  - $\sqrt{\text{Var}(b_k \mid X)}$ is the standard error of the sampling distribution of b_k.

11.2 Inference with OLS

- The purpose of inference is to use the sampling distribution

$$\frac{b_k - \beta_k}{\sqrt{\text{Var}(b_k)}} \sim N(0,1)$$

  to construct confidence intervals for β_k and conduct hypothesis tests about β_k.

- Given that the standard error of b_k must be estimated:

$$\widehat{\text{SE}(b_k)} = \sqrt{\widehat{\text{Var}(b_k)}}.$$

- The standardized estimator follows

$$\frac{b_k - \beta_k}{\widehat{\text{SE}(b_k)}} \sim t_{n-K},$$

  a t-distribution with $n - K$ degrees of freedom.

11.3 Confidence Interval Estimation for β_k

(1) Choose confidence level c.

  - Example: $c = 0.95$ gives a 95% confidence interval.

(2) Choose critical value κ from the t_{n-K} distribution:

$$\Pr(t > \kappa) = \frac{1-c}{2}.$$

(3) Construct the confidence interval:

$$\left[b_k - \kappa\widehat{\mathrm{SE}(b_k)}, \quad b_k + \kappa\widehat{\mathrm{SE}(b_k)}\right].$$

- Tradeoff:
  - Higher confidence $\Rightarrow$ larger κ $\Rightarrow$ wider interval.
  - Wider intervals are more likely to contain β_k, but are less precise.

11.4 Hypothesis Testing for β_k: t-Test

- Consider the hypotheses:

$$H_0 : \beta_k = q, \qquad H_1 : \beta_k \neq q.$$

- Calculate the t-statistic:

$$t = \frac{b_k - q}{\widehat{\mathrm{SE}(b_k)}}.$$

  - The t-statistic measures how far b_k is from q, relative to its sampling uncertainty.
  - If the null hypothesis is true,

$$t \sim t_{n-K}.$$

  - Large absolute values of t provide evidence against the null hypothesis.

11.5 Hypothesis Testing for β_k: P-Value

- Let $F_{n-K}(a)$ denote the CDF of a t distribution with $n - K$ degrees of freedom.
- For a two-sided test, the p-value is

$$\text{p-value} = 2\left[1 - F_{n-K}\left(|t|\right)\right].$$

- In R:

```
2 * (1 - pt(abs(tstat), df = n - K))
```

- Compare the p-value to the significance level α:
  - If p-value $< \alpha$, reject H_0.
  - If p-value $\geq \alpha$, fail to reject H_0.

11.6 Estimating the Variance of the OLS Estimates

- The variances of the OLS estimates are the diagonal elements of the variance-covariance matrix:

$$\mathrm{Var}(b \mid X) = (X'X)^{-1}X'\,\mathrm{E}(\boldsymbol{\varepsilon\varepsilon}' \mid X)X(X'X)^{-1}.$$

- Under homoskedasticity,

$$\mathrm{Var}(\varepsilon_i \mid X) = \sigma^2,$$

and non-autocorrelation,

$$\mathrm{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \qquad i \neq j.$$

- Therefore,
$$\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X) = \sigma^2 I_{n\times n},$$
and
$$\mathrm{Var}(b \mid X) = \sigma^2 (X'X)^{-1}.$$
- Thus, we only need to estimate σ^2.

11.7 Estimating the Variance of ε

- If we observed
$$\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n,$$
we could estimate the variance with
$$\widehat{\mathrm{Var}(\varepsilon)} = \frac{1}{n-1}\sum_{i=1}^{n}(\varepsilon_i - \bar{\varepsilon})^2.$$
- Instead, we observe the OLS residuals
$$e_1, e_2, \ldots, e_n.$$
which depend on the K estimated parameters in b.
- We estimate σ^2 with
$$s^2 = \frac{1}{n-K}\sum_{i=1}^{n} e_i^2.$$

11.8 Residual Sum of Squares

- Recall the residual-maker matrix
$$\mathbf{M}_X = I - X(X'X)^{-1}X'.$$
- Since
$$Y = X\beta + \boldsymbol{\varepsilon},$$
the OLS residuals are
$$\begin{aligned} e &= \mathbf{M}_X Y \\ &= \mathbf{M}_X(X\beta + \boldsymbol{\varepsilon}) \\ &= \mathbf{M}_X\boldsymbol{\varepsilon}, \end{aligned}$$
because $\mathbf{M}_X X = 0$.
- Therefore,
$$e'e = \boldsymbol{\varepsilon}'\mathbf{M}_X'\mathbf{M}_X\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'\mathbf{M}_X\boldsymbol{\varepsilon},$$
since $\mathbf{M}_X$ is symmetric and $\mathbf{M}_X^2 = \mathbf{M}_X$.

11.9 Using the Trace Operator

- From the previous slide,
$$e'e = \boldsymbol{\varepsilon}'\mathbf{M}_X\boldsymbol{\varepsilon}.$$
- The trace of a matrix is the sum of its diagonal elements.
- Since $e'e$ is a scalar,
$$e'e = \operatorname{tr}\left(\boldsymbol{\varepsilon}'\mathbf{M}_X\boldsymbol{\varepsilon}\right).$$
- A useful property of the trace is that matrices can be cycled:
$$\operatorname{tr}\left(\boldsymbol{\varepsilon}'\mathbf{M}_X\boldsymbol{\varepsilon}\right) = \operatorname{tr}\left(\mathbf{M}_X\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\right).$$
- The trace operator and expectation can be interchanged:
$$\begin{aligned}\mathrm{E}(e'e \mid X) &= \mathrm{E}\left[\operatorname{tr}\left(\mathbf{M}_X\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\right) \mid X\right] \\ &= \operatorname{tr}\left(\mathbf{M}_X\,\mathrm{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X\right]\right).\end{aligned}$$

11.10 s^2 is an Unbiased Estimate of σ^2

- Substituting
$$\begin{aligned}\mathrm{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X\right] &= \sigma^2 I_{n\times n}, \\ \mathrm{E}(e'e \mid X) &= \operatorname{tr}\left(\mathbf{M}_X\sigma^2 I_n\right) = \sigma^2\operatorname{tr}(\mathbf{M}_X).\end{aligned}$$
- Since
$$\begin{aligned}\mathbf{M}_X &= I_n - X(X'X)^{-1}X', \\ \operatorname{tr}(\mathbf{M}_X) &= \operatorname{tr}(I_n) - \operatorname{tr}\left(X(X'X)^{-1}X'\right) \\ &= n - \operatorname{tr}\left((X'X)^{-1}X'X\right) \\ &= n - \operatorname{tr}(I_K) = n - K.\end{aligned}$$
- Therefore,
$$\mathrm{E}(e'e \mid X) = \sigma^2(n-K) \quad \Rightarrow \quad \mathrm{E}\left(\frac{e'e}{n-K} \;\middle|\; X\right) = \sigma^2.$$

11.11 House Price Model

```
head(dat, 5)
```

```
##   price bdrms lotsize sqrft
## 1   300     4    6126  2438
## 2   370     3    9903  2076
## 3   191     3    5200  1374
## 4   195     3    4600  1448
## 5   373     4    6095  2514
```

```
lm(price ~ bdrms + I(sqrft/100) + I(lotsize/1000), data = dat)
```

```
##
## Call:
## lm(formula = price ~ bdrms + I(sqrft/100) + I(lotsize/1000),
##     data = dat)
##
## Coefficients:
##     (Intercept)            bdrms     I(sqrft/100)  I(lotsize/1000)
##         -21.770           13.853           12.278            2.068
```

11.12 Interpreting R-Output

```
mdl = lm(price ~ bdrms + I(sqrft/100) + I(lotsize/1000), data = dat)
summary(mdl)
```

```
##
## Call:
## lm(formula = price ~ bdrms + I(sqrft/100) + I(lotsize/1000),
##     data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -120.026  -38.530   -6.555   32.323  209.376
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -21.7703    29.4750  -0.739  0.46221
## bdrms            13.8525     9.0101   1.537  0.12795
## I(sqrft/100)     12.2778     1.3237   9.275 1.66e-14 ***
## I(lotsize/1000)   2.0677     0.6421   3.220  0.00182 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 59.83 on 84 degrees of freedom
## Multiple R-squared:  0.6724, Adjusted R-squared:  0.6607
## F-statistic: 57.46 on 3 and 84 DF,  p-value: < 2.2e-16
```

11.13 Model Variance-Covariance Matrix

```
vcov(mdl)
```

```
##                  (Intercept)        bdrms I(sqrft/100) I(lotsize/1000)
## (Intercept)      868.7780949 -162.3518384  -12.1358403      -0.4896188
## bdrms           -162.3518384   81.1827206   -6.2026557      -0.2683426
## I(sqrft/100)     -12.1358403   -6.2026557    1.7522896      -0.1128278
## I(lotsize/1000)   -0.4896188   -0.2683426   -0.1128278       0.4123256
```

```
diag(vcov(mdl))
```

```
##     (Intercept)           bdrms    I(sqrft/100) I(lotsize/1000)
##     868.7780949      81.1827206       1.7522896       0.4123256
```

```
sqrt(diag(vcov(mdl)))
```

```
##     (Intercept)           bdrms    I(sqrft/100) I(lotsize/1000)
##      29.4750419       9.0101454       1.3237407       0.6421258
```

11.14 Re-Constructing R-Output

```
Y = matrix(dat$price, ncol = 1)
X = cbind(rep(1, nrow(dat)), dat$bdrms, dat$sqrft/100, dat$lotsize/1000)
b = solve(t(X) %*% X) %*% (t(X) %*% Y)
Y_hat = X %*% b
resid = Y - Y_hat
s2 = sum(resid^2)/(nrow(X) - ncol(X))
diag(s2*solve(t(X) %*% X))
```

```
## [1] 868.7780949  81.1827206   1.7522896   0.4123256
```

```
diag(vcov(mdl))
```

```
##     (Intercept)           bdrms    I(sqrft/100) I(lotsize/1000)
##     868.7780949      81.1827206       1.7522896       0.4123256
```

```
sqrt(s2)
```

```
## [1] 59.83348
```

```
summary(mdl)$sigma
```

```
## [1] 59.83348
```

11.15 Example: Bivariate OLS

- For a constant and one predictor,

$$\hat{y}_i = b_1 + b_2 x_i.$$

- The matrices are

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}, \qquad X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}_{n \times 2}.$$

- The variance-covariance matrix is

$$\text{Var}(b \mid X) = \sigma^2 (X'X)^{-1} = \sigma^2 \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix}^{-1}.$$

11.16 Example: Bivariate OLS

- The variance-covariance matrix is

$$\text{Var}(b \mid X) = \sigma^2 \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix}^{-1}.$$

- Using the inverse of a 2×2 matrix,

$$\text{Var}(b \mid X) = \frac{\sigma^2}{n\sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2} \begin{bmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{bmatrix}.$$

- Therefore,

$$\text{Var}(b_2 \mid X) = \frac{\sigma^2 n}{n\sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2} = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

11.17 Example: Bivariate OLS

- For a constant and one predictor,

$$\hat{y}_i = b_1 + b_2 x_i.$$

- The variance of the coefficient on x is

$$\text{Var}(b_2 \mid X) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{\text{SST}_x}.$$

- The variance is determined by:

  1) The variance of the unobserved component of the model,

$$\text{Var}(\varepsilon_i \mid X) = \sigma^2.$$

  2) The total variation in the observed explanatory variable associated with the coefficient, SST_x.

11.18 Variance in the Multiple Regression Model

- In the linear regression model

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \varepsilon_i,$$

  focus on a single parameter β_k.

- Partition the model as

$$Y = X_1 \beta_1 + x_k \beta_k + \boldsymbol{\varepsilon},$$

  where X_1 contains all regressors except x_k.

- Using the residual maker $\mathbf{M}_1$,

$$\begin{aligned} \mathbf{M}_1 Y &= \mathbf{M}_1 x_k \beta_k + \mathbf{M}_1 \boldsymbol{\varepsilon} \\ &= x_k^* \beta_k + \boldsymbol{\varepsilon}. \end{aligned}$$

- Therefore,

$$\text{Var}(b_k \mid X) = \sigma^2 \left[(x_k^*)' x_k^*\right]^{-1} = \frac{\sigma^2}{\text{SST}_{x_k^*}}.$$

11.19 How X_1 Affects the Variance of b_k

- In
$$\text{Var}(b_k \mid X) = \frac{\sigma^2}{\text{SST}_{x_k^*}},$$
x_k^* is the residual from regressing x_k on X_1:
$$\hat{x}_k = X_1(X_1'X_1)^{-1}X_1'x_k, \qquad x_k^* = x_k - \hat{x}_k.$$
- Thus, $\text{SST}_{x_k^*}$ is the SSE from this auxiliary regression.
- Let $R^2_{k|X_1}$ denote the fraction of the variation in x_k explained by X_1:
$$R^2_{k|X_1} = 1 - \frac{\text{SST}_{x_k^*}}{\text{SST}_{x_k}}.$$

11.20 Rewriting the Variance of b_k

- In the model, let X_1 contain all regressors except x_k:
$$Y = X_1\beta_1 + x_k\beta_k + \boldsymbol{\varepsilon}.$$
$$\text{Var}(b_k \mid X) = \frac{\sigma^2}{\text{SST}_{x_k^*}}.$$
- From the previous slide, the denominator can be written as
$$\text{SST}_{x_k^*} = (1 - R^2_{k|X_1})\text{SST}_{x_k}.$$
- Therefore,
$$\text{Var}(b_k \mid X) = \frac{\sigma^2}{(1 - R^2_{k|X_1})\text{SST}_{x_k}}.$$

11.21 Determinants of the Variance of b_k

- In the linear regression model
$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \varepsilon_i,$$
- The variance of the coefficient on x_k is determined by
$$\text{Var}(b_k \mid X) = \frac{\sigma^2}{(1 - R_k^2)\text{SST}_{x_k}}.$$

1) The variance of the unobserved component of the model: σ^2.
2) The fraction of the variation in x_k not explained by the other variables in the model: $(1-R_k^2)$.
3) The total observed variation in x_k:
$$\text{SST}_{x_k} = \sum_{i=1}^{n}(x_{ik} - \bar{x}_k)^2.$$

11.22 What Improves the Precision of b_k

- Given
$$\text{Var}(b_k \mid X) = \frac{\sigma^2}{(1 - R_k^2)\text{SST}_{x_k}},$$
- The variance of b_k decreases when:
  1) There is more variation in x_k: $\uparrow \text{SST}_{x_k}$.
  2) The sample size increases: $\uparrow \text{SST}_{x_k}$.
  3) Relevant controls are added, reducing unexplained variation: $\downarrow \sigma^2$.

11.23 Multicollinearity Increases the Variance of b_k

- Given
$$\text{Var}(b_k \mid X) = \frac{\sigma^2}{(1 - R_k^2)\text{SST}_{x_k}},$$
- When x_k is highly correlated with the other regressors,
$$R_k^2 \approx 1 \quad \Rightarrow \quad 1 - R_k^2 \approx 0.$$
- If $R_k^2 = 1$, the model violates Assumption 2: full rank.
- Multicollinearity occurs when R_k^2 is close to one:
  - The model is close to violating the full-rank assumption.
  - The variance of b_k can become very large.
- But this effect can be offset by a smaller σ^2 or a larger SST_{x_k}.

11.24 Bias-Variance Trade-Off

- One measure of multicollinearity is the variance inflation factor:
$$\text{Var}(b_k \mid X) = \text{VIF}_k \frac{\sigma^2}{\text{SST}_{x_k}}, \qquad \text{VIF}_k = \frac{1}{1 - R_k^2}.$$
- In the LRM, we may need to add controls because omitting them would violate the Mean Independence assumption.
- These controls may be correlated with x_k:
  - $\uparrow R_k^2 \Rightarrow \uparrow \text{VIF}_k \Rightarrow \uparrow \text{Var}(b_k \mid X)$.
- Thus, reducing bias can come at the cost of reduced precision.
  - A reasonably large sample can help balance this trade-off.

11.25 Multicollinearity Can Cause Strange Results

- Warning Signs of Multicollinearity:
  - High R^2, but few statistically significant coefficients.
  - Small data changes produce large changes in coefficient estimates.
  - Coefficients have unexpected signs or implausible magnitudes.
- There is often no simple solution.
  - Dropping correlated variables can reduce multicollinearity, but may introduce omitted-variable bias.
- Be cautious when interpreting individual coefficients.

Lecture 12

OLS in Practice

12.1 Model Fit: R-squared

- Given the OLS estimates
$$b = (X'X)^{-1}X'Y,$$
- The OLS predictions and residuals are
$$\hat{Y} = Xb, \qquad e = Y - Xb.$$
- Total sample variation can be decomposed as
$$\text{SST} = \text{SSR} + \text{SSE}.$$
- R^2 measures the fraction of sample variation in y explained by X:
$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} \in [0, 1].$$

12.2 Interpreting R^2

- Since OLS minimizes the sum of squared errors (SSE), it also maximizes R^2.
- Consequently, adding regressors can never decrease R^2.
- When n is small relative to K, R^2 can overstate the model's explanatory power due to *overfitting*.
  - Overfitting occurs when the model captures random variation rather than meaningful relationships.

12.3 Adjusted R^2

- Adjusted R^2 is an alternative measure of fit that accounts for overfitting.
- Because OLS estimates K parameters, the residuals contain less random variation than appears.
  - Adjusted R^2 accounts for this by correcting for the degrees of freedom used in estimation.
- Conventional R^2:
$$R^2 = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$
- Adjusted R^2:
$$\text{adj-}R^2 = 1 - \frac{\sum_{i=1}^n e_i^2/(n-K)}{\sum_{i=1}^n (y_i - \bar{y})^2/(n-1)}.$$

12.4 Interpreting Adjusted R^2

- Adjusted R^2 can be negative.
- Adding a regressor can increase or decrease adjusted R^2.
  - An increase suggests the regressor is worth including.
- Adjusted R^2 uses estimates of the variance of y and ε:

$$\text{adj-}R^2 = 1 - \frac{\sum_{i=1}^{n} e_i^2/(n-K)}{\sum_{i=1}^{n}(y_i - \bar{y})^2/(n-1)} = 1 - \frac{\widehat{\text{Var}(\varepsilon)}}{\widehat{\text{Var}(y)}}.$$

  - This provides an estimate of out-of-sample fit.
- In contrast, R^2 measures in-sample fit.
  - Overfitting can cause R^2 to overstate out-of-sample fit.

12.5 Linear Probability Model

- Under the assumptions of the linear regression model,

$$\text{E}(y_i \mid x_i) = x_i'\beta.$$

- If y is binary, coded 0 or 1, then

$$\text{E}(y) = \Pr(y = 1) \cdot 1 + \Pr(y = 0) \cdot 0 = \Pr(y = 1).$$

- Thus, the conditional expectation function is a *linear probability model*:

$$\text{E}(y_i \mid x_i) = \Pr(y_i = 1 \mid x_i) = x_i'\beta.$$

- The OLS coefficient b_k estimates the change in the probability that $y_i = 1$ from a one-unit increase in x_{ik}, holding the other regressors fixed.

12.6 Linear Probability Model Example

- Suppose y indicates whether a worker belongs to a union:
  - $y = 1$ if the worker belongs to a union; $y = 0$ otherwise.
- Thus,

$$\text{E}(y) = \Pr(y = 1) = \Pr(\text{Union Member}).$$

- The estimated model is

$$\widehat{UNION} = 0.962 - 0.047ED.$$

- Multiplying b_2 by 100 gives the percentage-point (pp) change in probability from a one-unit increase in ED.
- One additional unit of ED is associated with a 4.7 pp decrease in the probability of union membership.

12.7 Interpreting Effects in Regressions with Logs

- We frequently apply the log transform to the data we are analyzing.
- For example, modeling the natural log of y instead of its level:

$$\log(y_i) = x_i'\beta + \varepsilon_i, \qquad y_i = \exp(x_i'\beta + \varepsilon_i).$$

- The marginal effect of x_{ik} on y_i is

$$\frac{\partial y_i}{\partial x_{ik}} = \exp(x_i'\beta + \varepsilon_i)\beta_k = y_i\beta_k.$$

- Expressing the marginal effect as a percentage of y_i:

$$\frac{\partial y_i}{\partial x_{ik}}\frac{100}{y_i} = 100\beta_k.$$

  - Thus, a one-unit increase in x_{ik} produces an approximate $100\beta_k$ percent change in y_i.

12.8 Interpreting Effects in Regressions with Logs

- Modeling the natural log of x_k instead of its level:

$$y_i = x_{i1}'\beta_1 + \log(x_{ik})\beta_k + \varepsilon_i.$$

- The marginal effect of x_{ik} on y_i is

$$\frac{\partial y_i}{\partial x_{ik}} = \frac{\beta_k}{x_{ik}}.$$

- Expressing the marginal effect for a 1% change in x_{ik}:

$$\frac{\partial y_i}{\partial x_{ik}}\frac{x_{ik}}{100} = \frac{\beta_k}{100}.$$

  - Thus, a 1% increase in x_{ik} changes y_i by approximately $\beta_k/100$ units.

12.9 Interpreting Effects in Regressions with Logs

- Modeling both y and x_k with natural logs instead of their levels:

$$\log(y_i) = x_{i1}'\beta_1 + \log(x_{ik})\beta_k + \varepsilon_i.$$

$$y_i = \exp(x_{i1}'\beta_1)x_{ik}^{\beta_k}\exp(\varepsilon_i).$$

- The marginal effect of x_{ik} on y_i is

$$\frac{\partial y_i}{\partial x_{ik}} = \beta_k \exp(x_{i1}'\beta_1)x_{ik}^{\beta_k - 1}\exp(\varepsilon_i) = \beta_k\frac{y_i}{x_{ik}}.$$

- Expressing the marginal effect as a percentage change in both variables (an elasticity):

$$\frac{\partial y_i}{\partial x_{ik}}\frac{x_{ik}}{y_i} = \beta_k.$$

  - Thus, a 1% increase in x_{ik} produces an approximate β_k% change in y_i.

12.10 Non-Constant Marginal Effects

- If x_k enters the model linearly and without interactions, its marginal effect is constant:

$$y_i = x_i'\beta + \varepsilon_i, \qquad \frac{\partial y_i}{\partial x_{ik}} = \beta_k.$$

- If x_{ik} enters nonlinearly, its marginal effect depends on the value of x_{ik}:

$$y_i = \beta_1 + x_{ik}\beta_2 + x_{ik}^2\beta_3 + \varepsilon_i, \qquad \frac{\partial y_i}{\partial x_{ik}} = \beta_2 + 2x_{ik}\beta_3.$$

- If x_{ik} is interacted with another variable, its marginal effect depends on the value of that variable:

$$y_i = \beta_1 + x_{ik}\beta_2 + x_{ij}\beta_3 + (x_{ik}x_{ij})\beta_4 + \varepsilon_i, \qquad \frac{\partial y_i}{\partial x_{ik}} = \beta_2 + x_{ij}\beta_4.$$

12.11 Causal Effect when x_{ik} is 0/1

- For a binary variable x_{ik},

$$y_i = x_{i1}'\beta_1 + x_{ik}\beta_k + \varepsilon_i,$$

  we measure the effect using a discrete change rather than a derivative.

- The causal effect is

$$\frac{\Delta y_i}{\Delta x_{ik}} = \beta_k,$$

  since $\Delta x_{ik} = 1$ when x_{ik} changes from 0 to 1.

- Under Mean Independence,

$$\beta_k = \mathrm{E}(y_i \mid x_{i1}, x_{ik} = 1) - \mathrm{E}(y_i \mid x_{i1}, x_{ik} = 0).$$

12.12 Bank Failures During the Great Depression

Richardson and Troost (2009), "Monetary Intervention Mitigated Banking Panics during the Great Depression," *Journal of Political Economy.*

- 1,000s of banks failed during the early 1930s.
- Fed policy differed across the 12 regional districts.
- Mississippi was split between two districts with different policy responses:
  - Atlanta Fed (6th District): more willing to lend to troubled banks.
  - St. Louis Fed (8th District): less interventionist.
- Did banks served by the more interventionist Atlanta Fed experience fewer failures than banks served by the St. Louis Fed?

12.13 Banks in Business Data

```
# Shows first 3 rows of data set
# bib is banks in business
head(bank, 3)
```

```
## # A tibble: 3 x 6
##    year month   day district   bib   bio
##   <int> <int> <int>    <dbl> <int> <int>
## 1  1929     7     1        6   141   141
## 2  1929     8     1        6   141   141
## 3  1929     9     1        6   141   141
```

```
nrow(bank)
```

```
## [1] 124
```

```
table(bank$district)
```

```
##
##  6  8
## 62 62
```

```
table(bank$year)
```

```
##
## 1929 1930 1931 1932 1933 1934
##   12   24   24   24   24   16
```

12.14 OLS with the `lm` Function

```
bank = bank %>%
  mutate(atlanta = +(district==6),
         crisis = +((year==1929 & month>=10) | (year>1929)),
         atlanta_crisis = atlanta*crisis)

lm(log(bib) ~ atlanta + crisis + atlanta_crisis, data = bank)
```

```
##
## Call:
## lm(formula = log(bib) ~ atlanta + crisis + atlanta_crisis, data = ba
##
## Coefficients:
##    (Intercept)         atlanta          crisis
##        5.12990        -0.18114        -0.26429
## atlanta_crisis
##        0.07761
```

```
# 100*-0.264 = -26.4% fewer banks in St. Louis during the crisis
# 100*(-.264 + 0.077) = -18.7% fewer banks in Atlanta during the crisi
# Atlanta's policy resulted in 100*0.077 = 7.7% fewer bank closures
```

12.15 Summarizing OLS

```r
mdl = lm(log(bib) ~ atlanta + crisis + atlanta_crisis, data = bank)
summary(mdl)
```

```
##
## Call:
## lm(formula = log(bib) ~ atlanta + crisis + atlanta_crisis, data = ba
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.17426 -0.13711 -0.02151  0.07952  0.27019
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     5.12990    0.07847  65.373  < 2e-16 ***
## atlanta        -0.18114    0.11097  -1.632  0.10525
## crisis         -0.26429    0.08044  -3.286  0.00133 **
## atlanta_crisis  0.07761    0.11376   0.682  0.49639
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1359 on 120 degrees of freedom
## Multiple R-squared:  0.2283, Adjusted R-squared:  0.209
## F-statistic: 11.83 on 3 and 120 DF,  p-value: 7.638e-07
```

Lecture 13

Extensions of the Linear Regression Model

13.1 Two Frequent Criticisms of the LRM

1) The OLS estimates are biased.
   - Important variables remain in the disturbance.
   - OLS cannot hold these variables fixed.
   - Mean Independence is violated.
2) The standard errors are incorrect.
   - Homoskedasticity is often unrealistic.
   - In some settings, non-autocorrelation is implausible.

13.2 Getting the Correct Standard Errors

- In the model
$$Y = X\beta + \boldsymbol{\varepsilon},$$
- We need $\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X)$ to construct standard errors.
- The classical LRM makes two assumptions:
  - Homoskedasticity: $\mathrm{Var}(\varepsilon_i \mid X) = \sigma^2$.
  - Non-autocorrelation: $\mathrm{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0$, $i \neq j$.
    $\implies \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X) = \sigma^2 I_{n\times n}$.
- If either assumption fails, conventional OLS standard errors are generally incorrect.
  - Incorrect confidence intervals and hypothesis tests.

13.3 Generalized Linear Regression Model

$$\begin{aligned} Y &= X\beta + \boldsymbol{\varepsilon} && \text{Linear} \\ \mathrm{E}(\boldsymbol{\varepsilon} \mid X) &= 0 && \text{Mean Independence} \\ \mathrm{Var}(\boldsymbol{\varepsilon} \mid X) &= \Sigma && \text{General Covariance} \end{aligned}$$

- Implications:
  1) OLS remains unbiased, but requires corrected standard errors.
  2) OLS remains LUE, but is no longer guaranteed to be MVLUE (more efficient linear unbiased estimators may exist).
  3) R^2 remains a valid measure of in-sample fit.

13.4 Robust Standard Errors

- Given the OLS estimator
$$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon},$$
- The variance of the estimator is
$$\begin{aligned}\operatorname{Var}(b \mid X) &= \operatorname{E}\left[(b-\beta)(b-\beta)' \mid X\right] \\ &= \operatorname{E}\left[(X'X)^{-1}X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X(X'X)^{-1} \mid X\right] \\ &= (X'X)^{-1}X'\operatorname{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X\right]X(X'X)^{-1}.\end{aligned}$$
- We can relax the homoskedasticity assumption by allowing
$$\operatorname{Var}(\varepsilon_i \mid X) = \sigma_i^2.$$
- This motivates *heteroskedastic-robust standard errors.*

13.5 Robust Standard Errors

- Under heteroskedasticity, the disturbance variance can differ across observations:
$$\operatorname{Var}(\varepsilon_i \mid X) = \sigma_i^2.$$
- Importantly, we do not need to estimate n additional parameters.
- In the generalized regression model:
$$\operatorname{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X) = \Sigma = \begin{bmatrix}\sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2\end{bmatrix}.$$
- We maintain the assumption of non-autocorrelation.

13.6 Huber-White Standard Errors

- Under heteroskedasticity, the variance of OLS is
$$\operatorname{Var}(b \mid X) = (X'X)^{-1}\operatorname{E}(X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X \mid X)(X'X)^{-1}.$$
- We only need to consistently estimate the middle term, not each σ_i^2, using squared OLS residuals:
$$\operatorname{E}(\widehat{X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X} \mid X) = X'\begin{bmatrix}e_1^2 & 0 & \cdots & 0 \\ 0 & e_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e_n^2\end{bmatrix}X.$$
- Apply a degrees-of-freedom correction for small samples:
$$\widehat{\operatorname{Var}(b \mid X)} = \frac{n}{n-K}(X'X)^{-1}X'\widehat{\Sigma}X(X'X)^{-1},$$
where $\widehat{\Sigma} = \operatorname{diag}(e_1^2, \ldots, e_n^2)$.

13.7 Revisiting the House Price Data

```
mdl = lm(price ~ bdrms + I(sqrft/100) + I(lotsize/1000), data = dat)
summary(mdl)
```

```
##
## Call:
## lm(formula = price ~ bdrms + I(sqrft/100) + I(lotsize/1000),
##     data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -120.026  -38.530   -6.555   32.323  209.376
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -21.7703    29.4750  -0.739  0.46221
## bdrms            13.8525     9.0101   1.537  0.12795
## I(sqrft/100)     12.2778     1.3237   9.275 1.66e-14 ***
## I(lotsize/1000)   2.0677     0.6421   3.220  0.00182 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 59.83 on 84 degrees of freedom
## Multiple R-squared:  0.6724, Adjusted R-squared:  0.6607
## F-statistic: 57.46 on 3 and 84 DF,  p-value: < 2.2e-16
```

13.8 Constructing Robust Standard Errors

```
Y = matrix(dat$price, ncol = 1)
X = cbind(rep(1, nrow(dat)), dat$bdrms, dat$sqrft/100, dat$lotsize/1000)
n = nrow(X)
K = ncol(X)
b = solve(t(X) %*% X) %*% (t(X) %*% Y)
Y_hat = X %*% b
resid =as.numeric(Y - Y_hat)
XXitX = solve(t(X) %*% X) %*% t(X)
var_b_HC = (n/(n-K))*(XXitX %*% diag(resid^2) %*% t(XXitX))
```

13.9 Comparing the Standard Errors

```
lmtest::coeftest(mdl)
```

```
##
## t test of coefficients:
##
##                  Estimate Std. Error t value  Pr(>|t|)
## (Intercept)     -21.77031   29.47504 -0.7386  0.462208
## bdrms            13.85252    9.01015  1.5374  0.127945
## I(sqrft/100)     12.27782    1.32374  9.2751 1.658e-14 ***
## I(lotsize/1000)   2.06771    0.64213  3.2201  0.001823 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lmtest::coeftest(mdl, vcov = var_b_HC)
```

```
##
## t test of coefficients:
##
##                 Estimate Std. Error t value  Pr(>|t|)
## (Intercept)     -21.7703    37.1382 -0.5862    0.5593
## bdrms            13.8525     8.4786  1.6338    0.1060
## I(sqrft/100)     12.2778     1.7725  6.9267 8.096e-10 ***
## I(lotsize/1000)   2.0677     1.2514  1.6523    0.1022
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

13.10 Robust Standard Errors the Easy Way

```
library(sandwich)
sqrt(diag(vcovHC(mdl, type = 'HC1')))
```

```
##     (Intercept)           bdrms    I(sqrft/100) I(lotsize/1000)
##       37.138211        8.478625        1.772533        1.251424
```

```
sqrt(diag(var_b_HC))
```

```
## [1] 37.138211  8.478625  1.772533  1.251424
```

```
# Robust-standard errors in one shot
mdl = lm(price ~ bdrms + I(sqrft/100) + I(lotsize/1000), data = dat)
lmtest::coeftest(mdl, vcov = vcovHC(mdl, type = 'HC1'))
```

```
##
## t test of coefficients:
##
##                 Estimate Std. Error t value  Pr(>|t|)
## (Intercept)     -21.7703    37.1382 -0.5862    0.5593
## bdrms            13.8525     8.4786  1.6338    0.1060
## I(sqrft/100)     12.2778     1.7725  6.9267 8.096e-10 ***
## I(lotsize/1000)   2.0677     1.2514  1.6523    0.1022
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

13.11 Testing for Heteroskedasticity

- Many tests for heteroskedasticity exist.
  - In practice, we often skip testing and just assume robust standard errors are needed.

- By construction, OLS residuals are uncorrelated with X since b is chosen so $X'e = 0$
  - The regressors cannot linearly predict the residuals.
- OLS imposes no restrictions on the variance of the residuals.
- If X predicts squared residuals, this provides evidence that

$$\mathrm{E}(\varepsilon_i^2 \mid X) \quad \text{depends on } X.$$

13.12 Breusch-Pagan (BP) Test

- Under homoskedasticity, the conditional variance is constant:

$$H_0 : \mathrm{Var}(\varepsilon_i \mid X) = \sigma^2.$$

- The BP test asks whether X predicts the squared disturbances:

$$\varepsilon_i^2 = \delta_0 + \delta_1 x_{i1} + \cdots + \delta_K x_{iK} + \nu_i.$$

- Since ε_i is unobserved, regress squared OLS residuals on X:

$$e_i^2 = \delta_1 + \delta_2 x_{i2} + \cdots + \delta_K x_{iK} + v_i.$$

- Under H_0, $\delta_2 = \cdots = \delta_K = 0$.
  - The auxiliary regression should have a small R^2.
- The LM test statistic is $LM = nR^2_{e^2} \sim \chi^2_{K-1}$.

13.13 BP Test on House Price Data

```
lmtest::bptest(mdl)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  mdl
## BP = 14.092, df = 3, p-value = 0.002782
```

13.14 Panel Data City Level Crime Data

```
dat = wooldridge::crime2
```

```
head(dat, 12)
```

```
##       pop    crmrte crimes unem officers year d87  area city
## 1  229528  74.65756  17136  8.2      326   82   0  44.6    1
## 2  246815  70.11729  17306  3.7      321   87   1  44.6    1
## 3  814054  92.93487  75654  8.1     1621   82   0 375.0    2
## 4  933177  89.97221  83960  5.4     1803   87   1 375.0    2
## 5  374974  83.61113  31352  9.0      633   82   0  49.8    3
## 6  406297  77.19476  31364  5.9      685   87   1  49.8    3
## 7  176496  88.94253  15698 12.6      245   82   0  74.0    4
## 8  201723  84.04099  16953  5.7      259   87   1  74.0    4
## 9  288446 108.17276  31202 12.6      504   82   0  97.3    5
## 10 331728 103.56377  34355  7.4      563   87   1  97.3    5
## 11 122768 136.89235  16806 13.9      186   82   0  55.3    6
## 12 142124 112.09225  15931  5.7      232   87   1  55.3    6
```

13.15 Panel Data

- Panel data contain repeated observations of the same units over time.
- Observations from the same unit share unobserved characteristics.
  - Disturbances are likely correlated within units.
  - Non-autocorrelation assumption may be violated.
- We must account for this correlation when estimating

$$\Sigma = \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X).$$

  - Some off-diagonal elements may be nonzero.
- This motivates *cluster-robust standard errors*.
  - Allow for both heteroskedasticity and within-cluster correlation.

13.16 Clustered Standard Errors with City Data

- The variance of OLS is

$$\mathrm{Var}(b \mid X) = (X'X)^{-1}\,\mathrm{E}(X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X \mid X)(X'X)^{-1}.$$

- Each city is observed twice. Allow disturbances to be correlated within cities:

$$\mathrm{E}(\widehat{X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X} \mid X) = X' \left[\begin{array}{cc|cc|cc} e_1^2 & e_1e_2 & 0 & 0 & 0 & 0 \\ e_2e_1 & e_2^2 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & e_3^2 & e_3e_4 & 0 & 0 \\ 0 & 0 & e_4e_3 & e_4^2 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & e_5^2 & e_5e_6 \\ 0 & 0 & 0 & 0 & e_6e_5 & e_6^2 \end{array}\right] X.$$

- Within-city correlations are allowed; between-city correlations are assumed zero.

13.17 Constructing Clustered Standard Errors

- Suppose the data contain G groups:
  - X_g contains the regressors for group g.
  - e_g contains the OLS residuals for group g.
- Estimate the middle term by summing within-group contributions:

$$\widehat{\mathrm{E}(X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X \mid X)} = \sum_{g=1}^{G} X_g' e_g e_g' X_g.$$

- Apply a small-sample correction:

$$\widehat{\mathrm{Var}(b \mid X)} = \frac{G}{G-1}\frac{n-1}{n-K}(X'X)^{-1}\left(\sum_{g=1}^{G} X_g' e_g e_g' X_g\right)(X'X)^{-1}.$$

- Allows arbitrary correlation within groups, but assumes independence across groups.

13.18 A Note on the Generalized Regression Model

- We estimate the disturbance covariance matrix under different assumptions:

$$X'\Sigma X = \mathrm{E}(X'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'X \mid X).$$

- We must impose some restrictions on the off-diagonal elements of Σ.
- Why not simply estimate Σ using ee'?

$$\widehat{\Sigma} = ee'$$

- This would imply

$$\begin{aligned}\widehat{\mathrm{Var}(b \mid X)} &= (X'X)^{-1}X'ee'X(X'X)^{-1} \\ &= 0,\end{aligned}$$

  - Because OLS imposes $X'e = 0$.

13.19 Correct Estimates. Wrong Standard Errors

```
mdl = lm(crmrte ~ unem + log(pop) + log(area) + d87, data = dat)
lmtest::coeftest(mdl)
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 135.68366   71.12129  1.9078  0.05972 .
## unem          0.61125    1.24131  0.4924  0.62366
## log(pop)     -3.93571    6.92240 -0.5685  0.57113
## log(area)     1.31348    4.93265  0.2663  0.79065
## d87           8.78174    8.18054  1.0735  0.28602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

13.20 Correct Estimates. Correct Standard Errors

```
suppressMessages(library(clubSandwich))
suppressMessages(library(lmtest))
mdl = lm(crmrte ~ unem + log(pop) + log(area) + d87, data = dat)
#Cluster vcovCR with "CR0", "CR1", "CR1S", ... differenet DF adjustments
coeftest(mdl, vcov = vcovCR(mdl, cluster = dat$city, type = 'CR1'))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 135.68366   92.90087  1.4605   0.1477
## unem          0.61125    1.13068  0.5406   0.5902
## log(pop)     -3.93571    8.64346 -0.4553   0.6500
## log(area)     1.31348    6.27584  0.2093   0.8347
## d87           8.78174    5.36993  1.6354   0.1056
```

13.21 Fixed Effects Models

- Establishing causal effects with the LRM is difficult.
  - Causal interpretation relies on mean independence.
- OLS holds included variables fixed, but not omitted variables.
- We cannot know for certain whether mean independence holds.
  - Our goal is to make the assumption *plausible*.

13.22 Causal Effects with Regression

- To estimate causal effects, we need to hold relevant factors fixed.
- Solution #1: Identify relevant variables and include them as controls.
- Solution #2: Estimate the model using only *within-subject variation*.
  - Holds fixed all observed and *unobserved* differences between subjects
  - This is the idea behind fixed effects.

13.23 Fixed Effects Model

- With repeated observations of the same units (cities, firms, people, etc.).
- Include indicator variables for each unit (fixed effects).

$$crmrte_{it} = \beta_1 + \beta_2 unem_{it} + \alpha_i + \varepsilon_{it}.$$

  - Estimate β_1, β_2, and a fixed effect α_i for each city.
- α_i captures *all* time-invariant characteristics of city i.
  - It acts as a sort of "super control."
  - Estimates the effect of unemployment holding city characteristics fixed.

13.24 Identification with Fixed Effects

- Consider the fixed effects model:

$$y_{it} = x_{it}'\beta + \alpha_i + \varepsilon_{it}.$$

- Fixed effects identify β using *within-unit variation*.
  - Compare changes in unemployment and crime within each city.
- The *between estimator* uses variation across units.
  - Compare average unemployment and crime across cities.
- OLS without fixed effects uses *total variation*.
  - Combines within- and between-city variation.

13.25 Identification of FE Model

- Write the fixed effects model in matrix form:

$$Y = X\beta + D\alpha + \boldsymbol{\varepsilon}.$$

- D is a matrix of group indicator variables:

$$D = \begin{bmatrix} \mathbb{1}_1 & 0 & \cdots & 0 \\ 0 & \mathbb{1}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbb{1}_G \end{bmatrix}.$$

  - $\mathbb{1}_g$ is a vector of ones, one for each observation in group g.
  - For example, if group 1 has 5 observations, $\mathbb{1}_1$ is a 5×1 vector.

13.26 Identification of FE Model

$$Y = X\beta + D\alpha + \boldsymbol{\varepsilon}$$

- The best predictor of y using only group indicators is the group mean.
- Partial out D:

$$\begin{aligned} M_D Y &= M_D X\beta + M_D D\alpha + M_D\boldsymbol{\varepsilon} \\ Y^* &= X^*\beta + \boldsymbol{\varepsilon}^* \\ y_{it} - \bar{y}_i &= (x_{it} - \bar{x}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i). \end{aligned}$$

- Fixed effects eliminate α_i by subtracting group means.

13.27 Corollaries of FE Identification

- *Corollary #1*: Groups with only one observation do not contribute to estimating β.
  - For these groups, $y_{it} - \bar{y}_i = 0$ and $x_{it} - \bar{x}_i = 0$.
  - Their residuals are zero for any b, so they contribute nothing to SSE.
- *Corollary #2*: Variables that do not vary within groups cannot be included.
  - If $x_{it} = \bar{x}_i$, then $x_{it} - \bar{x}_i = 0$.
  - Such variables are perfectly collinear with the fixed effects.

13.28 Estimating Fixed Effects Models

- Approach #1: Include group indicators and apply OLS.

$$Y = \begin{bmatrix} X & D \end{bmatrix} \begin{bmatrix} \beta \\ \alpha \end{bmatrix} + \boldsymbol{\varepsilon}.$$

  - Requires estimating many parameters when the number of groups is large.
- Approach #2: Partial out the fixed effects.
  - Subtract group means from y and x.

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

  - Computationally efficient with many fixed effects.

13.29 Fixed Effect Model with `fixest`

```
suppressMessages(library(fixest))
mdl = feols(crmrte ~ unem + log(pop) + log(area) + d87 | city,
            cluster = 'city',
            data = dat)
```

```
## The variable 'log(area)' has been removed because of collinearity (see
## $collin.var).
```

```
summary(mdl)
```

```
## OLS estimation, Dep. Var.: crmrte
## Observations: 92
## Fixed-effects: city: 46
## Standard-errors: Clustered (city)
##             Estimate Std. Error   t value  Pr(>|t|)
## unem         2.37681   0.822416  2.890038 0.0059084 **
## log(pop) -24.68326  36.676884 -0.672992 0.5043938
## d87       16.51526   5.162869  3.198854 0.0025283 **
## ... 1 variable was removed because of collinearity (log(area))
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 9.75759     Adj. R2: 0.771273
##                 Within R2: 0.203824
```

Lecture 14

Analysis of Time Series Data

14.1 Review: Cross-Sectional Data

- Cross-sectional data consist of observations from different sampling units (people, firms, cities, etc.).

$$\{(x_i, y_i)\}_{i=1}^{n}$$

- We typically assume observations are independently sampled from a population.
  - Each observation provides additional information about the population.
- To increase the sample size, we can collect additional observations.
  - Sample more units from the population.
  - In experiments, repeat the experiment with new subjects.

14.2 A Time-Series Process

- A time-series process, $\{y_t\}$, is a sequence of random variables indexed by time:

$$\{\ldots, y_{t-2}, y_{t-1}, y_t, y_{t+1}, y_{t+2}, \ldots\}$$

- Two important characteristics are:
  1) Time ordering: t, $t+1$, ...
  2) Potential correlation between observations over time:

$$\text{Cov}(y_{t+1}, y_t) \neq 0$$

14.3 Distinguishing Feature of Time-Series

- Consider observing national crime rates from 1992 to 2020:

$$crmrte_{1992}, crmrte_{1993}, \ldots, crmrte_{2020}$$

- While there are 29 observations over time, these data provide only a *single* realization of the time-series process.
  - We cannot independently resample the entire process.
- The central challenge for time-series analysis is learning about the underlying process from a single observed realization.

14.4 Time-Series Model

- A time-series model describes y_t in terms of explanatory variables x_t and a disturbance ε_t:

$$y_t = x_t'\beta + \varepsilon_t$$

- An important feature of time-series models is that past values can help explain current outcomes.
  - For example, x_t may include lagged explanatory variables, such as x_{t-1}, or lagged values of the dependent variable, such as y_{t-1}.
- Likewise, ε_t may depend on past disturbances, such as ε_{t-1}.
- These are all possible sources of dependence in a time-series process.

14.5 Two Goals of Time-Series Analysis

1) Time-series econometrics seeks to identify causal relationships using time-series data.
   - For example, how do interest rates affect GDP growth?
2) Time-series forecasting uses past and present data to predict future outcomes.
   - For example, what is the likely GDP growth rate next quarter?
   - Forecasting may involve predicting future disturbances, which is not generally a feature of cross-sectional prediction.

- The two approaches share some techniques but have important differences.

14.6 Estimation of Time-Series Models

- Time-series models are often estimated using ordinary least squares (OLS).
- Importantly, many time-series models do not satisfy the assumptions required for OLS to be unbiased.
- Instead, our central focus is on the conditions for OLS consistency.
  - In particular, the conditions under which a law of large numbers applies to time-series data.
- For statistical inference, we also need conditions under which a central limit theorem applies.
  - Constructing appropriate standard errors for OLS with time-series data is an important topic.

14.7 Finite Distributed Lag Model

- The effect of x on y may unfold over time.
- Time-series data allow us to study these dynamic effects.
  - This can reveal important relationships that are missed when we consider only contemporaneous effects.
- A finite distributed lag model takes the form:

$$y_t = \beta_1 + \beta_2 x_t + \beta_3 x_{t-1} + \beta_4 x_{t-2} + \varepsilon_t$$

- β_2 captures the contemporaneous effect of x_t on y_t.
- β_3 and β_4 capture the effects of past values of x on current y_t.

14.8 Example: GDP Growth and Interest Rates

- Consider a model of quarterly GDP growth, $gdpg_t$, and short-term interest rates, int_t:

$$\begin{aligned} gdpg_{t-1} &= \beta_0 + \beta_1 int_{t-1} + \beta_2 int_{t-2} + \varepsilon_{t-1} \\ gdpg_t &= \beta_0 + \beta_1 int_t + \beta_2 int_{t-1} + \varepsilon_t \\ gdpg_{t+1} &= \beta_0 + \beta_1 int_{t+1} + \beta_2 int_t + \varepsilon_{t+1} \end{aligned}$$

- β_1 captures the immediate effect of a change in interest rates on GDP growth.
- β_2 captures the effect in the following quarter.
- $\beta_1 + \beta_2$ captures the cumulative effect over two quarters.
- Under what conditions can we estimate this model with OLS?

14.9 Is OLS Unbiased?

- Consider the simpler model:

$$gdpg_t = \beta_0 + \beta_1 int_t + \varepsilon_t$$

- There are two important considerations when estimating this model using OLS.
- First, homoskedasticity and no serial correlation may not hold:

$$\text{Var}(\varepsilon_t) \neq \text{Var}(\varepsilon_{t-1}), \qquad \text{Cov}(\varepsilon_t, \varepsilon_{t-1}) \neq 0$$

  - We need standard errors that account for these possibilities.
- Second, mean independence may be plausible because interest rate decisions are not made in real time:

$$\text{E}(\varepsilon_t \mid int_t) = \text{E}(\varepsilon_t)$$

14.10 Review: OLS and Unbiasedness

- Recall the OLS estimator:
$$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$$
- To establish unbiasedness, we typically assume:
$$\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0$$
- This requires each disturbance to have conditional mean zero given all observations of the regressors:
$$\mathrm{E}(\varepsilon_i \mid x_1, \ldots, x_n) = 0$$
- This condition is called *strict exogeneity*.
- With randomly sampled cross-sectional data, independence across observations means we only need to establish:
$$\mathrm{E}(\varepsilon_i \mid x_i) = 0$$

14.11 Strict Exogeneity in Time-Series Models

- Strict exogeneity is difficult to establish in time-series models.
  - OLS estimators may therefore be biased in finite samples.
- Consider the model:
$$gdpg_{t-1} = \beta_0 + \beta_1 int_{t-1} + \varepsilon_{t-1}$$
$$gdpg_t = \beta_0 + \beta_1 int_t + \varepsilon_t$$
- Even if int_t does not respond to ε_t, it may respond to ε_{t-1}.
  - For example, the Fed may lower interest rates in response to unexpectedly weak GDP growth.
- Thus, $\mathrm{E}(\varepsilon_{t-1} \mid int_t) \neq \mathrm{E}(\varepsilon_{t-1})$ and strict exogeneity does not hold

14.12 Review: OLS and Consistency

- Showing that OLS is consistent requires less than strict exogeneity.
- Given the OLS estimator
$$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$$
- To show OLS is consistent, we appeal to the law of large numbers and require
$$\mathrm{E}\left(\frac{X'\boldsymbol{\varepsilon}}{n}\right) = 0$$
- This only requires $\mathrm{Cov}(x_i, \varepsilon_i) = 0$, assuming $\mathrm{E}(\varepsilon_i) = 0$.

14.13 Consistency in Time-Series Models

- Consider the model:

$$gdpg_{t-1} = \beta_0 + \beta_1 int_{t-1} + \varepsilon_{t-1}$$
$$gdpg_t = \beta_0 + \beta_1 int_t + \varepsilon_t$$

- Unlike unbiasedness, consistency does not require strict exogeneity.
- If a law of large numbers applies, OLS is consistent provided:

$$\text{Cov}(int_t, \varepsilon_t) = 0$$

- Consistency means $b_1 \xrightarrow{p} \beta_1$ as $T \to \infty$.
  - Somewhat paradoxically, time-series datasets often have relatively small sample sizes.

14.14 LLN and CLT for Time-Series Data

- LLN: Sample averages converge to population expectations.
- CLT: Sample averages have an approximately normal sampling distribution.
- The familiar versions of the LLN and CLT assume independent observations.
  - However, independence is not necessary for either result.
- For time-series data, alternative versions allow for correlated observations, which provide foundation for OLS consistency and statistical inference.
  - These typically require stationarity and sufficiently weak dependence.

14.15 Strict Stationarity

- A time-series process is *strictly stationary* if its joint distribution does not change over time.
- For any collection of periods and any shift h:

$$(y_{t_1}, y_{t_2}, \ldots, y_{t_k}) \stackrel{d}{=} (y_{t_1+h}, y_{t_2+h}, \ldots, y_{t_k+h})$$

- Implications:
  - The distribution of y_t is the same in every period.
  - The relationship between observations depends on their separation in time, not when they occur.
- A weaker condition, *covariance stationarity*, is often sufficient for time-series analysis.

14.16 Covariance Stationarity

- A time-series process is *covariance stationary* if:
  - Its mean is constant: $\mathrm{E}(y_t) = \mu$.
  - Its variance is constant: $\mathrm{Var}(y_t) = \sigma^2$.
  - Its autocovariance depends only on the lag:

$$\mathrm{Cov}(y_t, y_{t-h}) = \gamma_h.$$

- Covariance stationarity means the mean, variance, and autocovariance structure are stable over time.
- Our time-series regression models already impose some stability.
  - For example, we typically assume the coefficients β do not arbitrarily change over time.

14.17 Weak Dependence

- A time-series process is *weakly dependent* if observations become nearly independent as their separation in time increases.
- In particular, as $h \to \infty$:

$$\mathrm{Cov}(x_t, x_{t+h}) \to 0$$

  - Observations far apart in time become approximately uncorrelated.
- The rate at which dependence diminishes is important.
- With stationarity and sufficiently weak dependence, time-series versions of the LLN and CLT can apply.
  - These provide the foundation for OLS consistency and asymptotic normality.

14.18 Autoregressive Process of Order One: AR(1)

- Consider an AR(1) process:

$$y_t = \rho y_{t-1} + \varepsilon_t$$

  - The disturbances ε_t are assumed to be i.i.d.
- For a stationary AR(1) process, the autocorrelation at lag h is:

$$\mathrm{Cor}(y_t, y_{t-h}) = \rho^h$$

- If $|\rho| < 1$, the process is stationary and exhibits weak dependence:

$$\lim_{h \to \infty} \rho^h = 0$$

  - Observations become increasingly uncorrelated as their separation in time grows.

14.19 Estimating an AR(1) with OLS

- Consider an AR(1) process:

$$y_{t-1} = \rho y_{t-2} + \varepsilon_{t-1}$$
$$y_t = \rho y_{t-1} + \varepsilon_t$$

- Even though y_{t-1} is uncorrelated with the current disturbance ε_t, strict exogeneity fails:
  - The regressor y_{t-1} is correlated with the past disturbance ε_{t-1}.
- When $|\rho| < 1$, the stationary process exhibits weak dependence.
  - Thus, OLS is consistent even though it is biased in finite samples.
- Bias generally decreases as T increases.
- When $|\rho|$ is close to 1, dependence persists longer and finite-sample bias can be more pronounced.